

Physiological and genomic signatures of war and displacement: A comprehensive literature review and future directions

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ABSTRACT

There are now 108.4 million forcibly displaced people worldwide, many of whom endure adversities that result in trauma, toxic stress, and potentially, altered epigenetic development. This paper provides a comprehensive review of current literature on the biological signatures of war and forced migration among refugee populations. To consolidate evidence and identify key concerns and avenues for future research, we reviewed 36 publications and one article under review, published since 2000, most of which focused on refugees relocated in Europe and the Middle East. This body of work – including cross-sectional, observational, and experimental studies – reveals heterogeneous findings regarding human biological responses to war-related adversities and their associations with health outcomes. We conclude with four main observations, regarding why genomic and physiological biomarkers are valuable, what study designs advance understanding of causality and health-promoting interventions, how to prepare for ethical challenges, and why theoretical frameworks and research procedures need more detailed consideration in scientific publications.

1. Introduction

Profound global and political shifts have persistently molded patterns of migration, human health, early-life development, and intergenerational responses to war and forced displacement. Driven by violence, climate-related crises, and breaches of human rights, forced migration is now a major global challenge; by the end of 2022, there were approximately 108.4 million forcibly displaced people worldwide, including 62.5 million internally displaced people (IDPs), 35.3 million refugees, and 5.4 million asylum seekers (UNHCR, 2023). Crossing international borders in search of safety, many refugees endure harsh conditions, traumatic events, and toxic stress affecting biological and psychosocial wellbeing (Hynie, 2018; Jankovic-Rankovic et al., 2020; Meewisse et al., 2007; Panter-Brick et al., 2020b).

This review focuses on the physiological and genomic signatures of war trauma and forced migration among refugee populations. We begin with an overview of biological stress response systems, encompassing key elements such as the hypothalamic-pituitary-adrenal (HPA) axis, immune and cardiovascular system, inflammation processes, and the pivotal role of genomics in modulating these responses. Physiological biomarkers have been increasingly used in research with war-affected

and refugee populations (Panter-Brick et al., 2020b), to better understand pathways to health, resilience, and human flourishing (Panter-Brick, 2023). For its part, the field of genomics investigates the intricate dynamics of genes, their expression patterns, and their interactions with other genes and environmental factors, offering valuable insights into developmental processes and intergenerational trauma (Cleeren et al., 2011). In this paper, we review biomarker work undertaken with refugees and asylum-seekers (to the best of our knowledge, there are no comparable studies of internally-displaced people). We aim to review current knowledge and reflect upon why and how biological signatures of war and forced displacement are so variable within and between population groups. We highlight research challenges and gaps in current knowledge, underscoring the need for further investigation and exploration. In suggesting avenues for future research, we hope to advance theory, methods, and practice and provide evidence-based recommendations for working with forcibly-displaced populations.

All physiological and genomic biomarkers have the potential to offer valuable insights into human responses to adverse experiences (Barcelona de Mendoza et al., 2018; Decker et al., 2003; Dressler, 2005; McDade, 2002; Miller et al., 2007; Kuzawa and Sweet, 2009). However, each biomarker provides a specific window into what are systemic and

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integrated regulatory changes: we know that biological systems are highly interconnected, involving a dynamic interplay of genetic, epigenetic, development, and environmental factors (Guidi et al., 2021; Worthman and Panter-Brick, 2008). It is therefore essential to interpret biomarker data through the lens of temporal and developmental challenges, ecological and cross-cultural contexts, in order to better understand how underlying biological processes contribute to differential health outcomes in displaced populations.

Several theoretical models help to conceptually frame our understanding of stress response systems. For example, the *developmental origins of health and disease* (DOHaD) suggests that early life exposures, including social stressors, influence long-term health trajectories, with stress affecting the epigenome across generations (Gluckman et al., 2010; Kuzawa and Sweet, 2009). In turn, the *weathering hypothesis* calls attention to the long-term physiological wear-and-tear of social and environmental stressors, suggesting that chronic adversity contributes to health disparities through causing physiological dysregulation and aging (Geronimus et al., 2006). As we know, human development is inherently shaped by its context, leading individuals to perceive, respond to, and process various facets of their social, emotional, and

physical surroundings differently (Pluess, 2015). These differences manifest themselves in distinct psychoneuroendocrinological and/or behavioral response patterns, highlighting a complex interplay between context and individual development. This means that we need to pay careful attention to individual differences in environmental sensitivity, or *vantage sensitivity*, whereby some individuals are more or less vulnerable to the negative effects of adverse experiences but also more or less receptive to positive or nurturing experiences (Popham et al., 2021). The literature is also moving to a systems-level understanding of human health and development, with pathways of risk and resilience being analyzed at biological, social, and ecological level (Masten and Narayan, 2012; Panter-Brick, 2023). Multi-level perspectives are particularly relevant within the context of development during war and displacement-related traumatic experiences (Popham et al., 2021).

2. Overview of biological stress response systems

Experiences of war and forced migration are so taxing that human bodies respond by activating a number of physiological systems and, potentially, genetic and epigenetic responses (Box 1). At the forefront of

Box 1

Biological stress response systems: physiological and genomic biomarkers used in research on war and forced displacement.

Neuroendocrine system:

- Cortisol (blood, saliva, hair, and nail)
- Prolactin (blood)
- Epinephrin and norepinephrine (blood and saliva)
- Thyroxine (T4) and triiodothyronine (T3; blood)
- Dehydroepiandrosterone sulphate (DHEA-s; blood)

Cellular immunity:

- Epstein-Barr virus (EBV) antibodies (dried blood spots)
- Lymphocytes (T (CD2), B (CD20), NK (CD56, CD16), helper (CD4), and cytotoxic (CD8; blood)

Inflammation:

- C-reactive protein (CRP; blood and saliva)
- Interleukin-1 beta, Interleukin-18 (IL-1, IL-18; saliva)
- Interleukin-6, 8 (IL-6, IL-8; blood)
- Tumor necrosis factor alpha (TNF- α ; blood)
- Serum amyloid A (SAA; blood)

Cardiovascular system:

- Blood pressure (systolic and diastolic)

Neuroplasticity:

- Neurotrophic growth factor (BDNF)
- Nerve growth factor (NGF)
- Salivary alpha-amylase (sAA)

Genomics:

- DNA methylation (buccal swabs, saliva, blood)
- Monoamine oxidase A (MAOA) (buccal swabs)
- Serotonin transporter, 5-HTT (buccal swabs)
- Catechol-O-methyltransferase, COMT (buccal swabs)

stress responses is the *neuroendocrine* system. Specifically, the hypothalamic-pituitary-adrenal (HPA) axis governs the hormonal production of cortisol, with downstream impacts on metabolic, cardiovascular, immune, reproductive, and central nervous system activity (Cacioppo et al., 2015; Dickerson and Kemeny, 2004; Johnson et al., 1992; Miller et al., 2007; Koolhaas et al., 2011). In challenging situations, the body enhances energy release and cardiovascular function by increasing levels of circulating cortisol, which exerts neuroendocrine effects on the brain, affecting factors such as anxiety levels, attentiveness to stimuli, and memory processes (Yaribeygi et al., 2017). Given the critical regulating role of glucocorticoids on systems in the body, both reduced and elevated cortisol levels may result in physiological dysregulation, potentially leading to a number of adverse health outcomes (Johnson et al., 1992; Miller et al., 2007; McDade, 2000; Ursin and Eriksen, 2004).

Over time, dysregulation of the HPA axis can compromise immune function, specifically *cellular immunity* and general *inflammation*. Repeated exposure to stressful stimuli can lead to changes in cellular immunity and the reactivation of latent viruses, such as the Epstein-Barr virus (EBV; Glaser and Kiecolt-Glaser, 2005; McDade, 2007; Panter-Brick et al., 2020b). Thus elevated production of EBV antibody titers have been documented in, for example, contexts of rapid social change (McDade, 2002), daily stressors and political insecurity (Panter-Brick et al., 2008), and discrimination-related stress (McClure et al., 2010). Compromised immunity also leads to lymphocyte proliferative response and natural killer (NK) cell cytotoxic activity (Herbert and Cohen, 1993; Khedri et al., 2020). Lastly, chronic stress may suppress immune cell production and function, as shown in studies of bereaved caregivers (Esterling et al., 1994), natural disasters (Inoue-Sakurai et al., 2000), and work-related stressors (Bosch et al., 2009). A well-recognized marker of general *inflammation* in response to stress is C-reactive protein (CRP) (Pearson et al., 2003), an acute phase protein generated by the liver in response to pro-inflammatory cytokines. Elevated CRP levels are likely mediated by the neuroendocrine pathways, with consequences for cardiovascular disease risk and mortality (Li et al., 2017) predicting

conditions like coronary atherosclerosis (Shivpuri et al., 2012), cardiac ischemia (Pearson et al., 2003), and hypertension (Bautista et al., 2005).

Over-activation of stress response systems can also have negative effects on the *cardiovascular system* (Thayer, 2017). Chronic psychosocial stress has been associated with signs of cardiovascular dysregulation (Gravlee et al., 2005; Lucini et al., 2005; Sweet, 2010), indexed by high blood pressure, which greatly contributes to diminished life expectancy (Gravlee et al., 2005). Studies on discrimination and racism, social classification, social status, and caregiving has demonstrated associations between chronic psychosocial stressors and signs of autonomic dysregulation of the cardiovascular system (Gravlee et al., 2005; Lucini et al., 2005; Sweet, 2010).

Neuroplasticity is a crucial mechanism in the brain's ability to adapt, learn, and heal: it enables responses to challenges through rewiring and reorganizing its structure and function, ranging from molecular, synaptic, and cellular changes to more global network changes (Kays et al., 2012). Past research has shown that chronic stress and trauma-related disorders can have detrimental impacts on the structure and function of the brain (Hill and Martinowich, 2016; Leal et al., 2014). The limbic system, including structures like the amygdala and hippocampus, can show neuroplastic alterations that underlie dysfunctional trauma memories and adversely influence cognitive, emotional, and behavioral outcomes (Bremner, 2006). For instance, research employing magnetic resonance imaging (MRI) has found reduced hippocampal volume in individuals diagnosed with post-traumatic stress disorder (PTSD) and major depressive disorder (MDD), as well as associations between hippocampal volume and the severity and persistence of PTSD symptoms (Bremner et al., 2008; Woon et al., 2010).

Finally, *genetic* and *epigenetic* elements serve as important mediators influencing the interplay between genes and environments influencing health and development (Kuzawa and Sweet, 2009). Epigenetic factors encompass a diverse range of processes capable of modifying gene expression – in a heritable manner – without altering actual DNA sequences (Taki and de Melo-Martin, 2021). These include histone modifications, DNA methylation (DNAm), and RNA-mediated regulation (Al

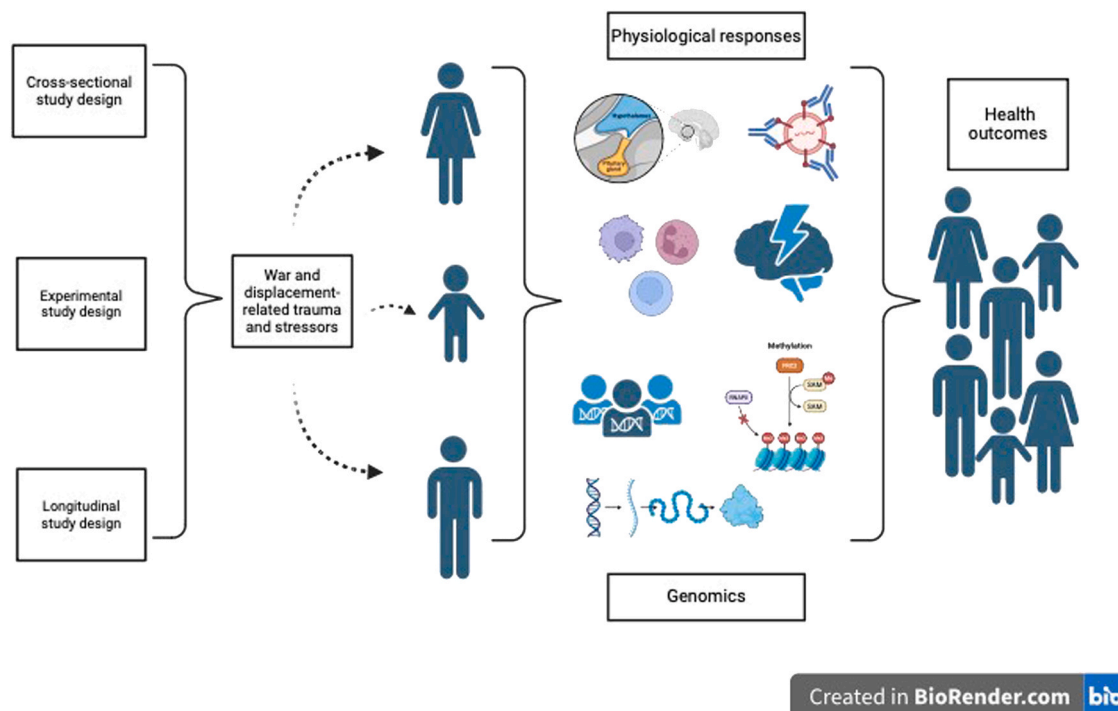


Fig. 1. Despite variations in study design, the findings demonstrate that war and displacement-related trauma and stressors elicit activation of multiple physiological systems (including neuroendocrine, immune, inflammation, and neuroplasticity) as well as genetic and epigenetic responses (such as DNA methylation). These responses have the potential to influence health outcomes.

Aboud et al., 2023; Martin et al., 2022), which modulate gene expression through regulatory molecules, affecting phenotypic development and health outcomes (Al Aboud et al., 2023; Taki and de Melo-Martin, 2021). For example, studies have shown that exposure to child abuse (Shields et al., 2016), prenatal stress (Ostlund et al., 2016), and war-related trauma (Mulligan et al., 2012) can influence distinct methylation patterns within genes associated with the HPA axis (Martin et al., 2022).

Focusing on the physiological and genomic markers of war trauma and forced migration (Fig. 1), this review utilizes a narrative methodology to discuss the malleability of human biological responses to war-related adversities and potential short and long-term adverse health outcomes. Results are presented according to study research design, in order to help evaluate the quality and strength of evidence from cross-sectional studies, randomized controlled trials, and observational studies.

3. Methods

We chose a narrative review methodology, an approach suited to synthesize knowledge where research is limited, or newly-evolving (Bryman, 2015). We included peer-reviewed studies that investigated the genetic, epigenetic, and physiological consequences of war trauma and forced displacement. We excluded non peer-reviewed and non-English language publications. Articles were identified using the following online electronic databases: Scopus, DOAJ, PubMed, and Google Scholar. We set publications dates as ranging between January 2000 and March 2024 (the first study measuring biomarkers in refugee populations was published in 2000). The key words and combination operators were: “forced migration AND health,” “refugees AND displacement,” “forced migration, refugees AND health,” “war, refugees AND biomarkers,” “war, refugees AND epigenetics,” “refugees AND physiology,” “refugees AND biomarkers,” “refugees and cortisol,” “refugees AND ebv,” and “refugees AND review AND meta-analysis.” The search resulted in a total 52 publications. We read, screened, and analyzed full texts. We excluded articles that analyzed mental health in the wake of war and forced displacement, but did not address genetics, epigenetics and/or physiology. Thirty-six publications and one article under review met inclusion criteria.

4. Results

4.1. Study characteristics

Study location and sample: Twelve studies were conducted in Europe, eleven the Middle East, five in Africa, and five in the USA (Tables 1 and 2). Most ($n=24$, out of 37) involved adult participants (18–70 years), rather than youth (8–19 years) or younger children (0–67 months). All but seven were published after 2015, echoing the unparalleled levels of displacement observed this past decade; in 2016, for example, the world witnessed profound refugee crises driven by conflicts, persecution, and instability in regions including Afghanistan, Syria, and Iraq, as well as North African countries, forcing millions of individuals to flee their homes in search of safety and refuge (UNHCR, 2019).

Research design: Most publications ($n=25$) were cross-sectional, one was randomized controlled trials, and eleven were longitudinal in design. All collected sociodemographic data and assessed traumatic experiences. They also included other measures of life adversity, resilience, sense of coherence, physical health, and/or substance use; nine incorporated anthropometric data such as body mass index (BMI). Mental health data consisted in self-reported survey measures of depression or anxiety, perceived stress, and post-traumatic-stress disorder (PTSD); two studies employed the structured clinical interview for DSM-IV assessment (Mattonet et al., 2023; Mewes et al., 2017).

Physiological biomarkers in war-affected populations: In measuring cortisol, one study opted for collection of saliva and urine samples

(Rohleder et al., 2004); two opted for blood samples (Hjort et al., 2021; Sabioncello et al., 2000); nine measured hair cortisol concentrations (HCC; Buchmüller et al., 2020; Dajani et al., 2018; Lembcke et al., 2020; May et al., 2024; Mewes et al., 2017; Panter-Brick et al., 2020b; Sierau et al., 2019; Shaheen et al., 2020; Smeeth et al., 2023a; Steudte et al., 2011); and three assessed nail cortisol concentrations (Gettler et al., 2024; Jankovic-Rankovic et al., 2020; Jankovic-Rankovic et al., 2022). Two studies assessed the hair level of the steroid hormone dehydroepiandrosterone (DHEA) (de Graaff et al., 2024; Schindler et al., 2019). Three studies (Gettler et al., 2024; Jankovic-Rankovic et al., 2022; Panter-Brick et al., 2020b) collected dried blood spots (DBS), because they provide a convenient and non-invasive method of measuring cortisol, EBV antibody levels, and/or CRP. One (Arnetz et al., 2020) examined brain derived neurotrophic growth factor (BDNF) and nerve growth factor (NGF), two primary neurotrophins crucial for neuroplasticity, and one (Thoma et al., 2012) the salivary enzyme alpha-amylase (sAA) a convenient and non-invasive indicator of the sympathetic nervous system (SNS) activity.

Genetic and epigenetic markers in war-affected populations: Two studies collected saliva DNA (Javanbakht et al., 2021; Smeeth et al., 2023b), four collected buccal DNA (Clukay et al., 2019a,b; Mattonet et al., 2023; Mulligan et al., 2022; Mulligan et al. under review), two measured blood DNA (Hjort et al., 2021; Rodney and Mulligan, 2014), and one measured skin conductance response (SKR; Javanbakht et al., 2021).

4.2. Cross-sectional studies

Overview. We identify four potential reasons that help explain heterogeneous findings in this body of work. The first stems from research design and small sample size (Box 2). Cross-sectional analyses have limited ability to make causal interpretations, observe change over time, and offer insights into intervention effectiveness and developmental processes. Study characteristics vary enormously, by participants' age, gender, living conditions, mental health profiles, and by choice of control groups, sample size, and representativeness. In the interpretation of biological data within war-affected refugee populations, the selection of an appropriate control group is pivotal (Table 3). For example, Rohleder et al. (2004) included both male and female participants, categorized into war-affected and control groups based on the presence or absence of a PTSD diagnosis, while Sabioncello et al. (2000) sampled only women, residing in precarious and stressful conditions, undifferentiated across PTSD or other mental health symptomology.

Sample size is a noteworthy limitation: for example, Steudte et al. (2011) compared ten individuals with PTSD with 17 traumatized controls, while Thoma et al.'s (2012) compared ten Bosnian refugees with PTSD with 11 controls. Such cross-sectional studies exemplify pioneering work, laying the foundation for advancements in the field and guiding future research.

Second, studies have captured different types of war- and displacement-related adversities. They often provide little explicit description of actual exposure to trauma events (severity, timing, and nature), and may or may not measure salutogenic factors (resilience, protective support, sense of coherence) or ongoing stressors likely to impact physiological outcomes. Importantly, it has been suggested that cortisol secretion follows a non-linear, two-staged pattern after traumatization; an initial rise in cortisol during the early post-traumatic phase, and a gradual transition over time to a state of hypocortisolism (Miller et al., 2007; Steudte-Schmiedgen et al., 2016). Studies reporting heightened cortisol levels have included refugees facing recent war-related adversities, while those reporting hypocortisolism have involved war-affected people who experienced trauma years or even decades ago. Studies also emphasize that we need to differentiate between past exposure to war and the ongoing stress of protracted displacement, both of which impact regulation of the HPA axis.

Third, studies have relied on different biological substrates to evidence changes in the biological stress response systems. Some seek to

Table 1
Physiological studies among war-affected refugee populations.

Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Arnetz et al. (2020)	Dysfunctional neuroplasticity in newly arrived Middle Eastern refugees in the U.S.: Association with environmental exposures and mental health symptoms	USA (Michigan)	Middle Eastern refugees from Syria (n = 40) and Iraq (n = 24) met in the Arab Community Center for Economic and Social Services (ACCESS) medical clinics	mean age of 37.63 years	1. Measures of trauma exposure (the Arabic version of the Harvard Trauma Questionnaire), environmental exposure (a 12-item revised version of a scale), PTSD symptoms (PTSD Checklist–Civilian version), depression (7- item depression subscale of the Hospital Anxiety and Depression Scale), and anxiety (21-item Beck Anxiety Inventory)	1. Neuroplasticity biomarkers (BDNF and NGF)	1. Self-reported sociodemographic data 2. Heavy metals concentration - manganese (Mn), cobalt (Co), nickel (Ni), arsenic (As), cadmium (Cd) and lead (Pb)	1. Investigating the associations between pre-displacement trauma, environmental war exposures, neuroplasticity and mental health outcomes in newly arrived refugees.	Cross-sectional study	1. Higher exposure to pre-displacement trauma was associated with more severe mental health symptoms but also with reporting higher exposure to war-associated environmental toxicants. 2. Higher levels of BDNF and NGF were associated with worse scores on validated scales for PTSD, depression, and anxiety, when controlling for pre-displacement exposures. 3. Refugees from Syria scored higher on the depression and anxiety scales. They also had significantly higher levels of manganese as compared to persons displaced from Iraq. 1. Lowered HCC in asylum seekers compared to immigrants. 2. A positive association between PTSD symptoms and HCC in internally displaced persons. 3. Significantly lowered cortisone concentration was found in internally displaced persons compared to immigrants. 4. A trend for higher DHEA levels was found in internally displaced persons compared to immigrants. 5. HCC was positively	1. Future studies on the assessment of self-rated measures of environmental exposures of concern, as well as determining levels of neurotrophins. 2. Future studies on the possible role of environmental exposures and its interaction with trauma exposures in the development of mental health disorders in refugees. 1. Longitudinal studies on war-affected populations are needed to address the question of whether and how alterations of the HPA axis functioning can set the stage for subsequent mental health problems. 2. Future studies should aim at disentangling the effects of early life stress and later traumatic events in war-affected populations. 3. Future studies should explore the relationship between physical diseases, mental disorders and
Buchmüller et al. (2020)	Exploring hair steroid concentrations in asylum seekers, internally displaced refugees, and immigrants	Middle East (Iraq) and Western Europe (Germany)	Internally displaced women living in refugee camps in Northern Iraq (n=14), Syrian asylum seekers (n=37), and Kurdish immigrants and asylum seekers (n=38) living in Germany	18–70 years	1. Measures of physical and mental health and PTSD symptoms (SF-12 PSS-10, and the 16 item PTSD-scale of the HTQ)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data 2. Self-reported measures of the experience of traumatic events affected by forced displacement (RATERS)	1. Assessing the usefulness of hair cortisol as a measure of chronic stress in populations affected by forced displacement.	Cross-sectional study	1. Lowered HCC in asylum seekers compared to immigrants. 2. A positive association between PTSD symptoms and HCC in internally displaced persons. 3. Significantly lowered cortisone concentration was found in internally displaced persons compared to immigrants. 4. A trend for higher DHEA levels was found in internally displaced persons compared to immigrants. 5. HCC was positively	(continued on next page)

Table 1 (continued)

Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Dajani et al. (2018)	Hair cortisol concentrations in war-affected adolescents: A prospective intervention trial	Middle East (Jordan)	Syrian refugee (n=411) and Jordanian non-refugee (n=322) in four urban centers in northern Jordan	12–18 years	1. Mental health data, including stress, human insecurity, PTSD symptoms (PSS, HI, CRIES)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data, including measures of lifetime trauma exposure, (21-item Trauma Event Checklist) 2. Self-reported resilience data (CYRM)	1. Analysis of hair cortisol as a marker of within- and between-person variation in adolescent responses to past trauma and current threats to wellbeing, in response to a brief intervention.	A randomized controlled trial (data collected in three time points - before the start of the psychosocial intervention trial (data collected in three time points - before the start of the psychosocial intervention (T1), at 2.8 months to capture the end of the 8-week intervention (T2), and at 11.3 months to capture a longer-term follow-up (T3).	associated with time since immigration in the control group. 1. Experiencing fear or insecurity was predictive of a trajectory of increased cortisol production. 2. The beneficial impacts of the 8-week stress-attunement intervention, leading to reductions in cortisol over time. 3. At the individual level, high-trauma youth exhibited hair cortisol concentrations that were likely dysregulated.	chronic stress in vulnerable populations. 1. In settings of conflict and extensive trauma, more research is needed to study both the impacts of psychosocial interventions and the etiology of hyper- and hypocortisolism.
de Graaff et al. (2024)	The associations of hair cortisol and DHEA with posttraumatic stress disorder symptoms in refugees	Europe (the Netherlands)	Syrian refugees (n=115)	18–69 years	1. Mental health data, including depression/anxiety, PTSD symptoms, (HSCL-25, PCL-5)	1. Stress-related biomarker data (hair CORT and hair DHEA)	1. Self-reported sociodemographic data 2. Self-reported traumatic events 3. Self-reported post-migration living difficulties (Post-Migration Living Difficulties checklist)	1. Investigating the complex relationships between the number of traumatic events and postdisplacement stressors, the 20 DSM-5 PTSD symptoms (clusters B to E), and HPA-axis hormones cortisol and DHEA assessed in hair samples in Syrian refugees with elevated psychological distress and impaired functioning.	Cross-sectional study (used the baseline pre-randomization data of a randomized controlled trial (RCT)	1. The number of traumatic events and post-displacement stressors were not related to cortisol and DHEA. 2. Cortisol and DHEA were differentially related to individual symptoms of PTSD, the number of traumatic events, and post-displacement stressors.	
Gettler et al. (2024)	Refugee health and physiological profiles in transitional settlements in Serbia and Kenya: Comparative evidence for effects of gender and social support	Eastern Europe (Serbia) and East Africa (Kenya)	Recently settled refugees (n=202) living in three asylum centers and refugees living in Kakuma refugee camp (n=163)	18–65 years	1. Measures of mental well-being (PSS, IES-R, RHS-15, FS) 2. Measure of perceived well-being (WHOQOL-Bref survey)	1. Stress-related biomarker data (fingernail CORT) 2. Immune biomarker data (EBV) 3. Inflammation biomarker data (CRP)	1. Self-reported sociodemographic data	1. Analyzing and comparing health, physiological, and psychosocial data from refugees in Serbia and Kenya, respectively, residing under divergent local conditions.	Cross-sectional study	1. Refugees residing in asylum centers in Serbia had higher nail cortisol levels than refugees residing in Kakuma in Kenya. 2. Refugees in Serbia had a lower likelihood of having elevated CRP compared to refugees at Kakuma, consistent with	1. Future psychobiological research regarding gender-based disparities in health would benefit from consideration of additional relevant physiological pathways that may be pertinent to different gendered dynamics.

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Table 1 (continued)

Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Ghandour et al. (2022)	Association Between Inflammatory Biomarkers and Mental Health Symptoms in Middle Eastern Refugees in the US	USA (Michigan)	Iraqi and Syrian refugees (n=64)	21–74 years	1. Measures of PTSD, anxiety, and depression (PLC-C, BAI, and HADS)	1. Inflammation biomarker data (blood IL-6 and IL-8)	1. Self-reported sociodemographic and economic data	1. Determining the association between war-related trauma exposure and the pro-inflammatory cytokines IL-6 and IL-8 and the association between symptoms of anxiety, depression, and PTSD symptom scores and the two inflammatory biomarkers.	Cross-sectional study	greater low-grade, chronic inflammation in the latter. 3. Refugee men had higher nail cortisol but lower EBV antibody titers and lower likelihood of having elevated CRP than did women. 1. A positive association between war-related trauma and all three mental health outcomes, anxiety, depression, and PTSD symptoms. 2. Age and war-related trauma exposure were significantly positively associated with IL-6 levels. 3. Depression was negatively associated with IL-6 levels. 4. Age and anxiety were positively associated with IL-8 levels. 1. Lower concentrations of IL-1 were associated with greater severity of post-traumatic stress symptoms. 2. A positive trend for greater IL-1 expression predicting higher anxiety and depression symptoms severity. 3. CRP also showed a negative relation to anxiety and depressive symptom severity. 4. Lower basal concentration of CRP predicted greater	1. Future studies should include a broader scope of inflammatory markers in refugees and their relationship to mental health and apply a prospective survey design to enhance our understanding of causal and temporal relationships. 1. Future research should place a greater focus on sex as a biological variable potentially driving relations between inflammation and psychological symptoms as well as possible diagnoses. 2. Future work is necessary to investigate inflammatory correlates of psychopathology by sex, in specific ethnic cohorts, and based on comorbidities.
Grasser et al. (2020)	Inflammation and Trauma-Related Psychopathology in Syrian and Iraqi Refugees	USA (Michigan)	Adult refugees (n=20 females, and n=16 males) from Syria (n=25) and Iraq (n=8)	18–65 years	1. Measures of PTSD, anxiety, and depression (PLC-C and HSCL-25)	1. Inflammation biomarker data (saliva CRP, IL β -1, and IL-18)	1. Self-reported sociodemographic data 2. Self-reported health	1. Examining the relation between inflammation and psychopathology in trauma-exposed Syrian and Iraqi refugees.	Longitudinal study		

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Table 1 (continued)

Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Jankovic-Rankovic et al. (2020)	Forced migration experiences, mental well-being, and nail cortisol among recently settled refugees in Serbia	Eastern Europe (Serbia)	Recently settled refugees (n=111) living in two asylum centers	18–55 years	1. Measures of mental well-being (PSS, IES-R, RHS-15)	1. Stress-related biomarker data (fingernail CORT)	1. Self-reported sociodemographic data 2. Semi-structured interviews	1. Examining the relationships between refugees' experiences during their dangerous journeys from their home countries, their current mental and physical health, and their CORT levels during their residence in two asylum centers in Serbia.	Cross-sectional study	anxiety symptoms severity. 1. Recently settled refugees who reported experiencing trauma during their arduous journeys had greater recent perceived stress and tended to exhibit higher CORT. 2. Refugees who experienced longer journeys reported higher recent perceived stress as well as poorer well-being and physical health. 3. Refugee women reported poorer well-being and physical health but also had lower CORT compared to men.	1. Longitudinal research tracking within-individual changes in refugees' biobehavioral profiles during various stages of forced migration and resettlement. 2. Further investigation of sex differences in health and physiological outcomes associated with refugee status. 3. Future research on women and men differences in health outcomes among refugees in more intermittent transitional stages of migration.
Jankovic-Rankovic et al. (2022)	Transient refugees' social support, mental health, and physiological markers: Evidence from Serbian asylum centers	Eastern Europe (Serbia)	Recently settled refugees (n=76) living in two asylum centers	18–50 years	1. Measures of mental well-being (PSS, IES-R, RHS-15, FS)	1. Stress-related biomarker data (fingernail CORT) 2. Immune biomarker data (EBV)	1. Self-reported sociodemographic data 2. Semi-structured interviews	1. Examining the associations between social support, mental health, and physiological markers among refugees in transitional migratory stages.	Cross-sectional study	1. Recently settled refugees, who reported lower social support, had poorer mental well-being and higher post-traumatic and recent perceived stress scores than other participants. 2. Refugees that had experienced longer journeys reported lower social support.	1. More research is needed among diverse refugee populations to better understand under what conditions psychological factors and social support may interrelate with immune markers, such as EBV antibody levels. 2. Longitudinal and intervention-based study designs are needed to disentangle key health-related drivers and social processes during transit following forced migration.
Lembcke et al. (2020)	Refugee mother-child dyads' hair cortisol, post-traumatic stress, and affectionate parenting	Western Europe (Germany)	Syrian and Iraqi refugee mothers with a child below the age of 5 years in the university (n=44 mother-child dyads)	Mothers: 21–45 years Children: 5–67 months	1. Preschool PTSD scale of CBCL (parents indicated responses) 2. PTSD scale (the fourth part of HTQ) 3. 12-item short form health survey (SF-12)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data, including SES 2. Parenting Interactions with Children Checklist of Observations Linked to	1. Investigating potential pathways between refugee mother-child dyads in a descriptive and explorative manner by analyzing self-report, observational, and hair cortisol data.	Cross-sectional study	1. High levels of PTSS in children who did not experience forced displacement themselves, but whether or not children had experienced flight was unrelated to	1. Future examination of the association between parenting practices and children's HCC in the context of forced displacement. Future studies need to be aware of potential

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Table 1 (continued)

Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
							Outcomes (PICCOLO)			their level of PTSS and HCC. 2. Children of typical HCC dyads had higher PTSS scores than children of atypical HCC dyads. 3. The direction of all associations with maternal affection, except children's PTSS, were dependent on the HCC group. Maternal affection was positively associated on a trend level with children's and mothers' HCC in typical dyads, whereas in atypical dyads those trends were reversed. 4. Mothers with higher scores of PTSS were more likely to show less maternal affection in typical dyads – the dyads in which children's PTSS were more pronounced. 5. Mothers' and children's levels of HCC were significantly correlated only in atypical HCC dyads. 6. There was a positive trend between children's HCC and PTSS. 7. Mothers' PTSS were linked to lower levels of maternal affection and higher child PTSS.	differential effects between and within samples when investigating family-related factors. They also need to take HCC synchrony as potential moderator into account. 2. Intervention strategies for refugee children should not only focus on coping with past traumatic experiences, but 3. More research is needed to understand the complex associations between traumatic events, PTSS, and HCC. 4. Future studies need to consider children's family-environment to a larger extent by including, for example, fathers and siblings, who are also important for children's development. 5. Longitudinal designs are needed to test the stability of HCC over time from early childhood until adulthood in order to fully understand underlying biological mechanisms of PTSS. Future studies should also control for intimate partner violence, childhood maltreatment, and other potential traumatic events to derive valid conclusions about war- and forced displacement. 6. Longitudinal studies need to investigate if a disrupted

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
May et al. (2024)	Hair hormone data from Syrian refugee children: Perspectives from a two-year longitudinal study	Middle East (Lebanon)	Child-caregiver dyads from Syria (wave1: n=1594 and wave 2: n=1000)	6–20 years		1. Stress-related biomarker data (hair CORT and hair DHEA)	1. Self-reported sociodemographic data 2. Self-reported health 3. Body mass index (BMI) 4. Hair-related variable 5. The effects of puberty (Pubertal Development Scale)	1. Promoting awareness about the usefulness and challenges associated with utilizing hair hormones.	Longitudinal study	1. Females had significantly higher cortisol concentrations, but lower testosterone and DHEA concentrations compared to males. 2. All hormone concentrations were substantially higher in adolescents compared to younger children. 3. Hormone concentrations tended to decrease with time since participants had left Syria, and were significantly higher in black colored hair compared to brown. 4. Elevated hair cortisol significantly predicted higher PTSD symptoms. 5. Children that had fled from Syria more than 12 months prior exhibited lower concentrations. 6. Elevated hair DHEA concentrations were significantly associated with more depression and anxiety.	endocrinological synchrony is a risk factor for, or a result of, PTSS. 1. Future studies on the biological effects on war-exposed refugees because of the persistence of hormone functioning miscalibration from childhood into adulthood, particularly regarding cortisol, which has shown various health-related consequences due to abnormal hormone regulation. 2. In future studies, researchers should consider collecting a comprehensive array of hair-related variables, including collecting hair sampling in a single season but longitudinally where possible to track hormone variability over time, while also organizing focus groups, carefully considering ethical implications, and ensuring thorough reporting of results to enable accurate comparison with other studies, alongside preregistration of the entire study for accountability and reproducibility.
Mewes et al. (2017)	Elevated hair cortisol concentrations in recently fled asylum seekers in comparison to permanently settled immigrants and non-immigrants	Western Europe (Germany)	Asylum seekers without PTSD (n=24), asylum seekers with PTSD (n=32), healthy permanently settled Turkish immigrants	20–49 years	1. Measures of traumatic events and PTSD symptoms (PDS + added items from the Harvard Trauma Questionnaire concerning traumatic events)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data 2. Structured Clinical Interview for DSM-IV (SCID-I) 3. Anthropometric data (BMI)	1. Comparing HCC, as a long-term marker for the endocrine stress response, of recently fled asylum seekers with PTSD to those without PTSD. 2. Comparing HCC of asylum seekers to HCC	Cross-sectional study	1. Relative hypercortisolism was found in recently fled asylum seekers. But, no difference in HCC between those with and without PTSD. 2. Relative hypercortisolism was	1. Future studies should complement peripheral measurements with molecular assessments. 2. Future studies should consider additional factors that possibly affect HCC (e.

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Munezero and Tomita, (2021)	Hypertension and Its Associated Mental Health Challenges Among Female African Refugees in Durban, South Africa	South Africa (Durban)	(n=24), and healthy nonimmigrant Germans (n=28) living in Germany	18–60+ years	1. Measures of exposure to psychological, physical, sexual abuse, and household dysfunction (the Adverse Childhood Experience Questionnaire) and depression (20-item Center for Epidemiologic Studies–Depression (CES-D))	1. Cardiovascular biomarker data (BP measures)	1. Self-reported sociodemographic data 2. Anthropometric data (BMI) 3. Physical activity (a self-reported five-question instrument) 4. Substance use (alcohol and cigarette or tobacco-related products)	of permanently settled immigrants and non-immigrant individuals.	Cross-sectional study	found in permanently settled Turkish immigrants.	g., physical activity). 3. Future studies may take a closer look at the longitudinal course of the biological alterations.
			Female refugees and asylum seekers (n=178) met in a nonprofit organization (NPO)					1. The high prevalence of hypertension among female refugees compared with previous national and international studies. 2. Exposure to adverse child experiences (ACE) and depression was associated with hypertension, independent of food insecurity, alcohol and smoking, obesity, and physical activity (exercise) status.		1. Future studies are needed that use diagnostic measures for both hypertension and depression and the associated mental health challenges.	
Panter-Brick et al. (2020b)	C-reactive protein, Epstein-Barr virus, and cortisol trajectories in refugee and non-refugee youth: Links with stress, mental health, and cognitive function during a randomized controlled trial	Middle East (Jordan)	Syrian refugees (n=446) and Jordanian non-refugees (n=371)	12–18 years	1. Mental health data, including psychosocial stress, human insecurity, psychiatric problems, PTSD symptoms (PSS, HI, AYM, SDQ, CRIES)	1. Stress-related biomarker data (hair CORT) 2. Immune biomarker data (CRP and EBV)	1. Self-reported sociodemographic data, including measures of adversity (21-item Trauma Events Checklist and 12-item Household Wealth Index of material goods) 2. Self-reported resilience data (CYRM) 3. Anthropometric data (BMI)	1. Evaluating the efficacy of a humanitarian intervention for youth affected by the Syria and Iraq crises in Jordan, Lebanon, Iraq, Syria, and Turkey targeting stress-alleviation. The first study to examine multiple biomarker trajectories in war-affected adolescents, in order to better evaluate the extent, timing, and malleability of the biological signatures of poverty, conflict, and forced displacement.	A randomized controlled trial (data collected in three time points before the start of the intervention (T1), just after the 8-week intervention (T2), and at 11.3 months to longer-term follow-up (T3)).	1. Three distinct trajectories for markers of inflammation (high, rising, and low CRP), different biomarkers two for cell-mediated immunity (high and low EBV), and three for hair cortisol (HCC hyper, medium, and hyposcretion), indicating substantial cohort heterogeneity and signaling differences in inflammatory processes, immune competence, and neuroendocrine stress across population sub-groups. 2. Robust associations between inflammatory response, neuroendocrine	1. Further examination of the etiology of gender differences. 2. The utility of the inflammation (high, rising, and low CRP), different biomarkers which are feasible to collect in humanitarian settings. 3. The impact of different types of interventions for reducing risks to healthy development among war-affected youth. 4. More work needed on different types of environmental adversity and reflect on what drives within-cohort differences in individual life history strategies. 5. More work needed with respect to elucidating the associations between
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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Pitts et al. (2016)	Determinants of altered intracellular endocrine-immune interplay in Bosnian war refugees suffering from PTSD	Western Europe (Germany)	Bosnian war refugees (n=52) with PTSD recruited from the Psychosocial Center (PSC) of Refugees in Duesseldorf and healthy German individuals (n=16) tested in the lab	22–56 years	1. Harvard Trauma Questionnaire (HTQ) 2. The revised Symptom Checklist-90 (SCL-90-R)	1. Glucocorticoid sensitivity (venous blood samples) 2. MRNA Expression	1. Self-reported sociodemographic data	1. Investigating the interplay between two key moderators, glucocorticoid receptor (GR α) and GR co-chaperone FKBP5, and their relation to GC sensitivity as well as contrasting the individual with the combined effect of GR α and FKBP5 on cortisol-immune signaling assessed by GC sensitivity of IL-6 producing cells in whole blood.	Cross-sectional study	stress, and BMI. 3. Associations between a rising CRP trajectory and perceived stress, and between HCC hypersecretion and perceived insecurity. 4. No evidence that the intervention had any impact on inflammation or immunocompetence for trial participants, relative to waitlisted controls. 1. Increased GC sensitivity of inflammatory cytokine production in whole blood in PTSD. 2. PTSD was in general associated with increased mRNA expression of both GR α and FKBP5 3. Elevated gene expression levels as well as highly correlated mRNA expression of GR α and FKBP5 were observed in the high GC sensitive PTSD group only. In contrast, patients in the low GC sensitive group showed reduced GR α mRNA expression, which further was unrelated to FKBP5 mRNA expression. 4. Marked expression upregulation of GR α and FKBP5 in PTSD patients. 5. GR α and FKBP5 were highly correlated in the healthy population,	adverse experiences and elevated CRP, especially for populations outside Western contexts. 6. Future work that will evaluate the usefulness of multiple biomarkers to track the differential effects of mental health problems and psychosocial stress on various physiological systems, over short and longer timescales, in responses to interventions. 1. The focus of the current study was on determinants of the altered intracellular endocrine-immune interplay observed in PTSD patients along the cortisol signaling pathway. Future studies may expand on the present findings by assessing the role of determinants along immune pathways as well as peripheral endocrine and immune markers, such as basal cortisol and plasma IL-6.

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Rohleder et al. (2004)	Hypocortisolism and Increased Glucocorticoid Sensitivity of Pro-Inflammatory Cytokine Production in Bosnian War Refugees with Posttraumatic Stress Disorder	Western Europe (Germany)	Bosnian refugees with PTSD diagnosis (n=12) and non-refugee control group (n=13) met at the Psychosocial Center of Refugees in Dusseldorf (PSC)	29–68 years	1. Measures of war-related stressful life events and PTSD symptoms (Harvard Trauma Questionnaire-traumatic experience section, International Diagnostic Checklist, SCL-90R, and 30-item section of the HTQ)	1.Stress-related biomarker data (saliva and urine CORT; urine epinephrine, and norepinephrine) 2. Physiological biomarker data (blood samples - GC sensitivity of inflammatory cytokine production)	1. Self-reported sociodemographic data 2. Self-reported wake-up times (CAR)	1. Investigating whether Bosnian war refugees with a PTSD diagnosis, compared with healthy control subjects, show hypocortisolism, whether these patients show an overactive inflammatory response, and whether GCs are effective in suppressing this inflammatory response.	Cross-sectional study	no association between these factors was observed in the PTSD group, indicating a linked expression only under non-PTSD physiological conditions. 6. mRNA expression of GR α and FKBP5 was highly correlated in the high GC sensitive subgroup, yet no relationship existed between these two variables in the low GC sensitive subgroup. 7. Significantly higher GR α expression levels were found in the high GC sensitive PTSD group compared to the low GC sensitive PTSD group. 1. A blunted free cortisol response to awakening was found in Bosnian war refugees with PTSD compared with control subjects without PTSD, as well as lower cortisol levels throughout the day. 2. PTSD patients displayed an increased sensitivity of pro-inflammatory cytokine production, together with higher cytokine producing capacity in peripheral blood monocytes. 3. The greatest differences in cortisol levels	1. More research is needed on whether the increased sensitivity reported in the study could be also found in nonimmunologic target tissues and whether this phenomenon can be reversed by successful therapeutic treatment.

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Sabioncello et al. (2000)	Immune, Endocrine, and Psychological Responses in Civilians Displaced by War	Eastern Europe (Croatia)	Female displaced persons (n=20) living in the refugee settlement and healthy control group (n=14)	18–39 years	1. Measures of neuropsychiatric and psychosomatic symptoms (COR-NEX2 test)	1. Stress-related biomarker data (serum concentrations of cortisol, prolactin, beta-endorphin, total T4, and T3) 2. Immune biomarker data (a complete white blood cell count; immunophenotyping of lymphocytes (T (CD2), B (CD20), NK (CD56, CD16), helper (CD4), and cytotoxic (CD8); lymphocyte proliferation, and polymorphonuclear phagocytic activity)		1. Investigating the immune, hormonal, and psychological status of multiply traumatized, displaced persons.	Cross-sectional study	between patients and control subjects were found in the early morning and throughout the day, whereas the evening levels showed smaller differences. 4. Hormones measured in urine collected overnight did not show any significant group differences. 1. Increased baseline cortisol, prolactin, and b-endorphin concentrations in displaced women and the positive correlation between cortisol and prolactin indicate elevated hypothalamic-pituitary-adrenal axis activity in chronic stress. 2. Displaced women exhibited depressive and anxiety disturbances. 3. A positive relation was found between personality traits and cortisol level. 4. The number of CD8 lymphocytes was not altered, the relative proportion of these cells was decreased, while the number of NK cells was increased. 5. The principal finding was the increase in lymphocyte subsets expressing activation markers (CD71, CD23, and HLA-DR) and in proliferating lymphocytes.	

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Schindler et al. (2019)	Victims of War: Dehydroepiandrosterone Concentrations in Hair and Their Associations with Trauma Sequelae in Palestinian Adolescents Living in the West Bank	Middle East (West Bank)	Female adolescents (n=92)	11–16 years	1. Trauma exposure and PTSD symptomatology (UCLA PTSD Reaction Index for DSM IV, Child) 2. Measures of trait anxiety (the State-Trait Anxiety) 3. Measures of the presence and severity of depressive symptomatology (The Center for Epidemiological Studies Depression Scale) 4. Measures of the resilience factor SoC, (SoC Scale and the SoFC Scale. 5. An assessment of sleeping problems, (Mini Sleep Questionnaire)	1. Stress-related biomarker data (hair CORT and hair DHEA)	1. Self-reported sociodemographic and hair-related data 2. Body mass index (BMI) 3. Pubertal stage	1. Examining hair DHEA and hair cortisol levels in the context of chronic traumatic stress and coping in adolescents living in the West Bank.	Cross-sectional study	1. Trauma-exposed individuals had significantly higher DHEA levels and lower cortisol/DHEA ratios compared to those in the control group. 2. DHEA and cortisol/DHEA ratio were found to be linked to trauma load and timing, but not to coping.	1. Future studies need focusing on male participants when assessing DHEA in the context of traumatic stress.
Shaheen et al. (2020)	Victims of war—Psychoendocrine evidence for the impact of traumatic stress on psychological well-being of adolescents growing up during the Israeli–Palestinian conflict	Middle East (West Bank)	Palestinian adolescents (n=70 males; n=163 females)	11–16 years	1. Measures of posttraumatic stress disorder (PTSD) and traumatic experiences (UCLA PTSD Reaction Index) 2. Measures of depression (CES-DC), 3. Measures of habitual anxiety (STAI-Y-Trait) 4. Measures of a sense of coherence (SoC; SOC-13 and SOFC)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data 2. Self-reported health	1. Assessing the impact of ongoing traumatic stress caused by a violent conflict on the psychological and psychoneuroendocrine functioning in a cohort of Palestinian adolescents.	Cross-sectional study	1. HCC was significantly elevated in the PTSD subgroup compared to the subgroup not exposed to any traumatic events, with trauma-exposed individuals in between. 2. HCC was further associated with typical sequelae of traumatic stress. 3. SoC was inversely related to self-reported psychopathology, as well as to HCC in the trauma group.	1. Further examination of the associations between trauma, SoC, and psychoendocrine markers is required to get closer to possible diagnostic applications. 2. Future studies should address gender differences by comparing hijab- and nonhijab-wearing female participants regarding their HCC
Sierau et al. (2019)	Hair cortisol, lifetime traumatic experiences and psychopathology in	Western Europe (Germany)	Unaccompanied refugee minors from Syria and Afghanistan (n=91)	14–19 years	1. Measures of lifetime incidence of traumatic events (LEC-5) 2. PTSD symptoms (PLC-5) 3. Depression	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data 2. Anthropometric data (BMI)	1. Investigating associations between long-term cortisol output, assessed through HCC, lifetime traumatic events and psychological	Cross-sectional study	1. HCC is negatively related to two different measures of lifetime traumatic experiences, the total number of traumas	1. Future work will need to examine the association between HCC and emotional problems. 2. Future research providing more

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
					symptoms (PHQ-9) 4. Symptoms of anxiety (GAD-7 and SSS-8) 5. Self-reported emotional and conduct problems (SDQ)			symptoms in a sample of unaccompanied refugee minors.		and the number of different trauma types. 2. HCC is positively related to self-reported emotional problems, but not with any of the other symptoms (PTSD, depression, anxiety, somatic symptoms, conduct problems. 1. HCC was positively associated with the number of war-related events.	detailed longitudinal analyses and insights into HCC trajectories is needed.
Smeeth et al. (2023a)	War exposure, post-traumatic stress symptoms and hair cortisol concentrations in Syrian refugee children	Middle East (Lebanon)	Child-caregiver dyads from Syria (n=1000)	8–16 years	1. War exposure (WEQ) 2. The quality of living conditions (perceived refugee environment index -PREI) 3. Child-reported PTSD symptoms (CPSS)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic data 2. Self-reported health 3. Body mass index (BMI) 4. Pubertal stage	1. Examining the relationships between war exposure, the quality of current living conditions, HCC and PTSD in Syrian refugee children and adolescents.	Longitudinal study	1. HCC was positively associated with the number of war-related events. 2. Time since war exposure is inversely related with HCC. 3. Older children exhibited higher HCC as well as greater war exposure. 4. Children with greater PTSD symptom severity exhibit elevated HCC. Cross-sectional findings: 1. SAA and hsCRP were lower in PTSD individuals. Longitudinal findings: 1. SAA and hsCRP had lower mean values in subjects with PTSD at baseline assessment compared with non-PTSD. 2. In the PTSD subjects, the difference from first to last follow-up of SAA and the difference in DHEAs were negatively correlated. 3. Self-rated symptoms of PTSD	1. Future work will need to investigate how more recent experiences alter trauma-related cortisol secretion.
Søndergaard et al. (2004)	The inflammatory markers C-reactive protein and serum amyloid A in refugees with and without posttraumatic stress disorder	Northern Europe (Sweden)	Iraqi refugees (n=86; n=32 women and n=54 men)	18–48 years	1. Measures of general health (GHQ-28) 2. Measures of PTSD symptoms (IES-22) 3. Measures of anxiety and depression (HLS-25)	1. Inflammation biomarker data (highly-sensitive CRP, serum amyloid A (SAA) 2. Stress-related biomarker data (blood CORT and dehydroepiandrosterone sulfate (DHEA-s))	1. Self-reported sociodemographic data 2. Self-reported health	1. Examining whether the inflammatory markers C-reactive Protein (CRP) and Serum amyloid A (SAA) differed among subjects with and without post-traumatic stress disorder (PTSD) in a group of refugees with a high prevalence of trauma exposure and PTSD.	Cross-sectional/Longitudinal study	1. SAA and hsCRP had lower mean values in subjects with PTSD at baseline assessment compared with non-PTSD. 2. In the PTSD subjects, the difference from first to last follow-up of SAA and the difference in DHEAs were negatively correlated. 3. Self-rated symptoms of PTSD	1. Future studies need to gather information on recent or intercurrent infections in order to eliminate a possible source of confounding. 2. Future work should study different populations of PTSD subjects.

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Steudte et al. (2011)	Increased cortisol concentrations in hair of severely traumatized Ugandan individuals with PTSD	East Africa (Uganda)	PTSD individuals (n=10) and traumatized controls (n=22)		1. Measure of perceived stress (PSS-4) 2. Measure of lifetime traumatic events 3. Measure of depression (HSCL and M.I.N.I.)	1. Stress-related biomarker data (hair CORT)	1. Self-reported sociodemographic and economic data 2. Body mass index (BMI)	1. Utilizing hair cortisol analysis to obtain an index of cumulative cortisol secretion in PTSD patients and traumatized controls under naturalistic living circumstances.	Cross-sectional study	and difference in log-transformed SAA were negatively correlated. 1. Significantly increased hair cortisol levels in individuals with PTSD compared to traumatized controls. 2. A positive association between the cumulative exposure to traumatic stressors and hair cortisol concentrations.	1. Future research examining participants repeatedly following traumatic events is needed to determine the specific influence of the time interval since traumatization. 2. Future research should investigate groups which are characterized by comparable levels of traumatic load in order to examine the specific impact of PTSD diagnosis on cortisol. 3. Future studies should make comparisons between PTSD, traumatize individuals, and healthy non-traumatized controls. 4. Future research should thus combine hair cortisol analysis with other established measures of HPA axis activity to further enhance the understanding of neuroendocrinological changes in a severely traumatized PTSD sample.
Thoma et al. (2012)	Altered salivary alpha-amylase awakening response in Bosnian War refugees with posttraumatic stress disorder	Western Europe (Germany)	Bosnian refugees with PTSD (n=10) and control group (n = 11)	29–68 years	1. Measures of war-related stressful life events and PTSD symptoms (HTQ, SCL-90R, and HTQ)	1. Biomarker of SNS activity (salivary alpha-amylase (sAA) 2. Physiological biomarker data (blood samples - GC sensitivity of inflammatory cytokine production - IL-6 and TNF- α)	1. Self-reported sociodemographic data	1. Assessing diurnal profiles of salivary alpha-amylase in PTSD, evaluate the usefulness of salivary alpha-amylase as an additional tool for studying stress related disorders, and to better understand SNS alterations in PTSD.	Cross-sectional study	1. PTSD patients showed sAA awakening response profiles opposite of those seen in healthy individuals, such that PTSD patients showed an increase instead of a sharp decrease in sAA values after awakening. 2. sAA secretion	(continued on next page)

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Author	Title	Geographical area	Sample and Location	Age	Self-Reported Mental Health Measures	Physiological biomarker data	Other	Purpose of study	Study design	Main findings	Future directions
Yamin et al. (2021)	The Development of Posttraumatic Stress Disorder and Depression Symptoms in Iraqi Refugees: Associations with Acculturation and C-reactive Protein	USA (Michigan)	Male Iraqi refugees (n=53)	18–57 years	1. Measures of pre-displacement trauma exposure (HTQ) 2. Measures of acculturation (MAAS) 3. Measures of PTSD symptoms (PLC-C) 4. Measures of depressive symptoms (HADS)	1. Immune biomarker data (blood CRP)	1. Self-reported sociodemographic data 2. Anthropometric data (BMI)	1. Examining how contextual factors in the host country, particularly acculturation and a biomarker of inflammation, CRP, are associated with the development of mental health, such as PTSD and depression	Longitudinal study	patterns to be positively associated with self reports of psychiatric symptoms of PTSD and anxiety. 3. Higher sAA wake-up responses is associated with higher GC sensitivity of inflammatory cytokine production. 1. Acculturation was inversely related to PTSD and depression symptoms at 2-year follow-up. 2. Higher CRP levels, but not acculturation, were associated with greater PTSD and depression symptoms at 2-year follow-up. 3. Both lower acculturation and higher CRP levels were associated with greater depression symptoms.	1. Future research should focus on the development and testing of interventions to reduce inflammation, especially for vulnerable populations such as refugees. 2. Future studies should examine the role of acculturation, inflammatory markers, and symptoms in larger samples of refugees from other nations and should include women. 3. Future research should measure changes in acculturation or CRP longitudinally. 4. Future work should include other factors, such as postmigration economic and employment factors as well as discrimination, when looking into the development of psychiatric symptoms in refugees.

understand chronic stress exposure, through hair and nail, others seek to understand diurnal regulation or dysregulation, in blood and saliva. Fourth, scholars have been interested in different mental health outcomes, and used different indicators of mental health burden. Much of the literature has focused on PTSD, depression, and anxiety, with often scant attention to the effectiveness and relevance of psychometric questionnaires across cultures and age groups. To tackle such methodological challenges, it is crucial to meticulously document research practices and types of data, let alone ensure ethical conduct and community engagement in the field, going beyond the accepted scientific practices of detailing laboratory and statistical procedures.

4.2.1. Neuroendocrine system

Cortisol in saliva, urine, and blood. We begin with two studies that examined the associations between cortisol levels and experiences of war trauma and forced displacement. Both compared war refugees with healthy controls. The first study compared a small sample ($n=12$) of Bosnian war refugees with PTSD and a control group of healthy individuals ($n=5$ Bosnian refugees without PTSD and $n=8$ German laboratory staff). The authors (Rohleder et al., 2004) collected 12-hour urine samples for assessment of urinary cortisol, epinephrine, and norepinephrine, in addition to saliva samples over the course of two consecutive days, at three distinct time points. They found no significant differences in urinary cortisol, epinephrine, and norepinephrine levels between the PTSD refugees and controls. However, as compared to healthy controls, the refugees with PTSD exhibited a blunted free salivary cortisol response to awakening, along with consistently lower cortisol levels throughout the day. The second study focused on Croatian refugees ($n=20$) and a control group of healthy Croatian researchers ($n=14$) unexposed to war. The research team (Sabioncello et al., 2000) found that refugees exhibited higher serum cortisol levels compared to the healthy control group.

Hair cortisol concentrations. Three studies have identified elevated HCC among displaced populations. The first compared three groups ($n=108$): asylum seekers with and without PTSD, Turkish immigrants, and healthy non-immigrant individuals in Germany (Mewes et al., 2017). The authors found that the asylum seekers exhibited a 42% higher HCC compared to the nonimmigrant group, while the permanently settled immigrants showed 23% lower HCC compared to the non-immigrant group. The second study also compared three groups ($n=233$), all residing in the West Bank: Palestinian adolescents given an initial PTSD diagnosis, trauma-exposed individuals without PTSD, and adolescents with no exposure to traumatic events (Shaheen et al., 2020). The research team documented a significant increase in HCC among the former group. They also found an inverse relationship between sense of coherence (SoC) and self-reported psychopathology, as well as between SoC and HCC in the trauma-exposed group. The third study, conducted in Northern Uganda (Steudte et al., 2011), found evidence that individuals with PTSD ($n=10$) exhibited higher HCC in compared to traumatized controls ($n=17$). All participants had experienced relatively recent war-related trauma and were still residing in precarious conditions.

The opposite pattern of findings have been reported by three other studies that found lower HCC levels among displaced populations. The first (Buchmüller et al., 2020) compared three female refugee groups ($N=89$): asylum seekers from Syria, internally displaced people (IDP) in Iraq, and Kurdish immigrants who resettled to Germany. The authors found that Syrian asylum seekers living in Germany exhibited lower HCC compared to Kurdish immigrant controls. PTSD symptoms and perceived stress in IDPs were linked to elevated cortisol levels. The second (Lembcke et al., 2020) examined HCC among refugee mother-child dyads from Iraq and Syria, categorizing them on the basis of median mother-child HCC difference (0.42); typical HCC dyads ($n=18$) had levels in the range of 0.42–2.92, while atypical HCC dyads ($n=17$) had lower levels in the range of -1.40 – 0.36 . Moreover, in typical dyads, refugee children exhibited higher post-traumatic stress

symptoms (PTSS) levels. The third study (Sierau et al., 2019) focused on 91 male unaccompanied refugee minors (URM) from Syria and Afghanistan, residing in residential group homes in Germany. The authors found inverse correlations between HCC and trauma exposure. Additionally, HCC was positively associated with self-reported emotional problems (using data from the Strength and Difficulties Questionnaire), but not with self-reported data for PTSD, anxiety, and/or somatic symptoms.

Fingernail cortisol concentrations. To-date, only three studies have evaluated cortisol levels in fingernails among both refugees in more intermittent transitional stages of forced migration and refugees residing in camps for an extended period of time (Gettler et al., 2024; Jankovic-Rankovic et al., 2020; Jankovic-Rankovic et al., 2022). They found that recently-settled refugees residing in Serbian asylum centers, who reported experiencing trauma and higher psychosocial stress, had higher cortisol levels (Jankovic-Rankovic et al., 2020). Individuals reporting lower levels of social support tended to exhibit elevated cortisol (Jankovic-Rankovic et al., 2022), suggesting that social support serves as a key resource for promoting the recovery, resilience, and overall health of refugees during transitional migratory phases. Additionally, Jankovic-Rankovic and her team noted that the choice of collecting nails instead of hair samples was influenced by women's hesitancy to unveil their hijab in public or in the presence of a male translator, as well as men's preference for having very short hair, shaved close to the scalp. Gettler et al. (2024) combined and analyzed physiological and psychosocial data from refugees ($N=365$) residing in Kenya and Serbia, respectively, with divergent temporal (length of stay) and socio-ecological conditions. They found that refugees residing in asylum centers in Serbia ($n=202$) had higher nail cortisol levels than refugees residing in Kakuma in Kenya ($n=163$). Additionally, the research team observed gender disparities, with refugee men exhibiting higher nail cortisol in comparison to refugee women.

Hair DHEA concentrations. To-date, only two studies have evaluated DHEA levels in hair among refugee populations. One study (Schindler et al., 2019) compared two groups of female adolescents ($n=92$), aged 11–16, all residing in the war-affected West Bank territory: trauma-exposed individuals and individuals without traumatic experiences. The research team assessed DHEA levels in hair along with HCC levels in both groups. They found that trauma-exposed individuals had significantly higher DHEA levels and lower cortisol/DHEA ratios compared to those in the control group. In addition, DHEA and cortisol/DHEA ratio were found to be linked to trauma load and timing, but not to coping. The other study (de Graaff et al., 2024) stands out for being the only one to utilize network analysis in investigating the association between HPA-axis hormones cortisol and DHEA and individual symptoms of PTSD among adult Syrian refugees ($n=115$; aged 18–69) based in the Netherlands. Although the study did not use a comparison or control group, the researchers found that HCC and DHEA levels were differently linked to the 20 PTSD symptoms outlined in the DSM-5. However, neither hormone was correlated with the number of traumatic events experienced or post-displacement stressors.

4.2.2. Cellular immunity

In war-affected contexts, findings related to cellular immunity are also inconsistent. The study by Sabioncello et al. (2000) found a proliferation of lymphocytes in female Croatian refugees, relative to healthy controls. It also found that serum cortisol levels were positively correlated with numbers of helper T cells (CD4) and activated lymphocytes (CD71). Gettler et al. (2024) found that refugee men had lower EBV antibody titers than did refugee women. For their part, Jankovic-Rankovic et al. (2022) in Serbia found no significant association between social support and EBV antibodies. Not enough is known about EBV as a reliable indicator of physiological responses to war trauma and forced displacement, given that extant literature has predominantly engaged Euro-American populations such as adult caregivers or university students (Glaser et al., 1985; Kiecolt-Glaser et al.,

Table 2
Genomic studies among war-affected refugee populations.

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Clukay et al. (2019a)	Association of MAOA genetic variants and resilience with psychosocial stress: A longitudinal study of Syrian refugees	Middle East (Jordan)	Adolescent Syrian refugees (n=399; n=221 males and n=178 females)	12–18 years	1. Measures of trauma exposure (Traumatic Events Checklist), resilience (CYRM-12), and psychosocial stress (PSS-14) 2. Measures of human distress (HD) and Human Insecurity (HI) 3. Measures of depression and anxiety (AYMH) 4. Measures of behavioral and emotional mental health difficulties (SDQ) 5. Measures of PTSD symptoms (CRIES-8)	1. Buccal DNA	1. Self-reported sociodemographic data	1. Testing for association of MAOA variants with different measures of psychosocial stress and mental health in a population of Syrian refugee youth who have experienced high levels of war-related trauma. 2. Investigating the protective aspects of a contextually specific measure of resilience that measures individual and collective strengths, resources, and capabilities.	Longitudinal study	1. No significant associations between MAOA and the five tested outcomes (PSS, HD, AYMH, SDQ, or CRIES-8) in females. 2. In males associations were only significant for symptoms of psychosocial stress (PSS), but not for AYMH or SDQ. 3. MAOA was significantly associated with PSS over time, with MAOA-L males who had sharper reductions in PSS levels over time as compared to high MAOA-H males. 4. There was no interactive effect of MAOA and trauma exposure on PSS. 5. While resilience was strongly associated with PSS, no significant interactive effects between MAOA variants and resilience on PSS was observed.	1. Future studies should study additional genes in order to fully understand the impact of childhood trauma and resilience on adult behaviors, particularly in refugee populations with exposure to high levels of trauma. 2. Work with a larger, more heterogeneous sample is needed to more fully evaluate the effects of MAOA, trauma exposure, and resilience on psychosocial stress and mental health measures, particularly in non-Western contexts with high levels of trauma exposure.
Clukay et al. (2019b)	FAAH, SLC6A4, and BDNF variants are not associated with psychosocial stress and mental health outcomes in a population of Syrian refugee youth	Middle East (Jordan)	Adolescent Syrian refugees (n=400; n=221 males and n=179 females)	12–18 years	1. Measures of trauma exposure (Traumatic Events Checklist),	1. Buccal DNA	1. Self-reported sociodemographic data	1. Testing for associations between three candidate genes, lifetime trauma exposure, and	Longitudinal study	1. No association between variants in these candidate genes and psychosocial	1. Further investigations into the genetics of psychosocial stress and mental health, particularly with

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
					resilience (CYRM-12), and psychosocial stress (PSS-14) 2. Measures of human distress (HD) and Human Insecurity (HI) 3. Measures of depression and anxiety (AYMH) 4. Measures of behavioral and emotional mental health difficulties (SDQ) 5. Measures of PTSD symptoms (CRIES-8)			resilience levels with six measures of psychosocial stress and mental health in a longitudinal sample of Syrian refugee youth who re-settled in Jordan.		stress or mental health outcomes.	respect to trauma and resilience over time, are needed.
Hjort et al. (2021)	Intergenerational effects of maternal post-traumatic stress disorder on offspring epigenetic patterns and cortisol levels	Eastern Europe (Kosovo)	118 mothers and 120 children	30–59 years	1. Measure of PTSD symptoms (CAPS-IV)	1. Stress-related biomarker data (blood CORT) 2. Blood DNA	1. Self-reported sociodemographic data and lifestyle factors	1. Investigating the association between maternal post-traumatic stress disorder during pregnancy and offspring DNA methylation and cortisol levels.	Cross-sectional study	1. Offspring of mothers with PTSD during pregnancy had higher cortisol levels and differential methylation at a number of genes, but none of the methylation differences reached epigenome-wide corrected significance levels. 2. Maternal PTSD was associated with differential methylation at a number of stress-related candidate genes, including NR3C1, HTR3A and BDNF, and DNA methylation	1. Further work is required in larger studies that ideally follow women from the time of pregnancy, where retrospective recall bias of PTSD is minimized, and which measure DNA methylation in neonates. 2. Longitudinal epigenetic studies focusing family therapy intervention is also needed to examine if its effectively possible to reverse the epigenetic modifications with PTSD.

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Javanbakht et al. (2021)	Biological and Environmental Factors Affecting Risk and Resilience among Syrian Refugee Children	Michigan (USA)	Child-caregiver dyads from Syria (n=1359)	7–17 years	1. Measure of PTSD, anxiety and depressive symptoms (PCL-C, HSCL-25, SCARED, LEC, UCLAPSTD, MFQ) 2. HTQ 3. Measures of living conditions (LDQ) 4. Measures of Perceived Residential Environment Quality Indicators and Neighborhood Attachment 5. Measures of pubertal development 6. Measure of somatic symptoms (SSS-8) 7. Measures of household food access and security	1. Saliva DNA 2. Skin conductance response (SKR)	1. Self-reported sociodemographic data	1. To determine the course of anxiety and PTSD symptoms across years 1 and 2 post-war trauma. 2. To determine the SCR correlates of symptom course in children. 3. Examine epigenetic age acceleration in refugee children and its association with symptom changes. 4. Examine genome-wide DNA methylation changes over the course of three years post migration and their association with symptom course in children. 5. Determine the SCR correlates of the differences in symptoms severity between refugee children and non-refugee immigrant children.	Longitudinal study	at a number of genes was associated with cortisol levels. 1. Maternal anxiety and depression symptom severity were associated with anxiety symptom severity in the child. 2. Skin conductance was highly responsive to trauma narrative. 3. DNAm age was highly correlated with chronological age. 4. Multiple CpGs associated after FDR correction at 5%.	
Kertes et al. (2016)	Prenatal Maternal Stress Predicts Methylation of Genes Regulating the Hypothalamic–Pituitary–Adrenocortical System in Mothers and Newborns in the Democratic Republic of Congo	Central Africa (Democratic Republic of Congo)	24 mother–newborn dyads		1. Stressful life experiences, chronic stress, and war-related trauma (oral history) 2. Measure of peri-natal stress	1. Blood DNA (maternal venous blood, placental tissue, and umbilical cord blood)	1. Self-reported general and reproductive health	1. Examining the effects of prenatal stress exposure on DNA methylation in genes regulating the HPA axis and associations of methylation with birth weight.	Cross-sectional study	1. A total of 18 CpG sites were significantly associated with chronic stress or war trauma in one or more tissue. 2. Among the CpG sites associated with prenatal stress,	1. Future research should examine the relations of global (gene-level) and specific (CpG-level) methylation with HPA axis activity throughout development.

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
										11 were linked to chronic stress, and 14 were associated with war trauma, with six sites predicted by both stressors. 3. Considering CpG sites surviving FDR correction, those with prior evidence of association, or located at a transcription factor binding (TFB) region, chronic stress predicted methylation in CRH and NR3C1, explaining 16%–25% of the variance. 4. War trauma predicted methylation in all four genes. Effects were observed in all four genes in placenta and maternal blood, while in cord blood, significant associations with stressors were found in CRH and NR3C1 only. 5. Three potentially relevant CRH sites were identified in cord blood. In placenta, reduced methylation at three NR3C1 sites. In maternal blood, one	

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Kertes et al. (2017)	BDNF methylation in mothers and newborns is associated with maternal exposure to war trauma	Central Africa (Democratic Republic of Congo)	24 mother–newborn dyads		1. Stressful life experiences, chronic stress, and war-related trauma (oral history) 2. Measure of peri-natal stress	1. Blood DNA (maternal venous blood, placental tissue, and umbilical cord blood)	1. Sel-reported general and reproductive health	1. Examining associations of prenatal exposure to maternal stress and BDNF methylation at CpG sites across the BDNF gene.	Cross-sectional study	NR3C1 site showed significant methylation associated with war trauma. 6. Except for one FKBP5 site, where higher methylation was associated with chronic stress and war trauma in placenta and with chronic stress in maternal blood. 7. Regarding birth weight, methylation at four sites across two genes predicted BW, explaining 55% of the variance in birth weight. 1. Maternal experiences of war trauma and chronic stress were associated with BDNF methylation in umbilical cord blood, placental tissue, and maternal venous blood. 2. Associations of maternal stress and BDNF methylation showed high tissue specificity. 3. The majority of significant associations were observed in putative transcription factor binding regions.	1. Future research on BDNF methylation should follow a two-pronged approach: (a) identifying loci meeting strict statistical significance criteria ($p < .05$ and $q < .25$) irrespective of prior knowledge, and (b) considering loci with unadjusted $p < .05$ that are strategically located at putative transcription factor binding sites or supported by prior evidence of association for further exploration (continued on next page)

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Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Mattonet et al. (2023)	No association between war-related trauma or PTSD symptom severity and epigenome-wide DNA methylation in Burundian refugees	East Africa (Tanzania)	230 family triads - 317 participants were included: 110 children (49% female) and 207 caregivers (53% female)	7–15 years (children) 19–74 (caregivers)	1. Measure of war-related trauma load (Violence, War, and Abductee Exposure Scale) 2. Measure of PTSD (childre - PTSD Reaction Index for DSM-5, and caregivers - PTSD Check List for DSM-5) 3. Measure of caregivers' substance use (ASSIST)	1. Buccal DNA	1. Self-reported sociodemographic data 2. Structured clinical Interview	1. Investigating associations between war-related trauma, PTSD, and altered DNAm patterns in Burundian refugee families.	Cross-sectional study	1. In the EWAS, analyses on the WGCNA-modules and the post-hoc sensitivity analyses, neither war- related trauma load nor PTSD symptom severity were significantly associated with the epigenome-wide DNAm levels of the retrieved CpGs nor the module's EigenCpGs from WGCNA, neither for the children nor for the caregivers.	in studies examining BDNF expression. 1. Future studies with larger sample size should account for such interaction effects to be able to identify potential significant effects on DNAm profiles. 2. Future studies would benefit from a longitudinal approach to account for temporal dynamics in DNAm signatures established and changed by pre-clinical, clinical, and post-clinical conditions of trauma exposure and PTSD. 3. Future large-scale EWAS with underrepresented conflict-affected African populations are needed for a better understanding on effects of war-related trauma and PTSD on DNAm in African individuals.
Mulligan et al. (2012)	Methylation changes at NR3C1 in newborns associate with maternal prenatal stress exposure and newborn birth weight	Central Africa (Democratic Republic of Congo)	24 mother–newborn dyads		1. Stressful life experiences, chronic stress, and war-related trauma (oral history) 2. Measure of peri-natal stress	1. Blood DNA (maternal venous blood, placental tissue, and umbilical cord blood)	1. Sel-reported general and reproductive health	1. Examining if maternal prenatal stress may result in epigenetic at NR3C1 in newborn that are related to birth outcomes.	Cross-sectional study	1. War stress have the most substantial effect, explaining 35% of the variance in birth weight. 2. The strongest correlation was found between war stress and newborn	

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Mulligan et al. (2022)	Novel GxE effects and resilience: A case: control longitudinal study of psychosocial stress with war-affected youth	Middle East (Jordan)	Syrian refugee (n=417) and Jordanian non-refugee youth (n=306); (n=415 males and n=308 females)	12–18 years	1. Measures of trauma exposure (Traumatic Events Checklist), resilience (CYRM-12), and psychosocial stress (PSS-14) 2. Measures of human distress (HD) and Human Insecurity (HI) 3. Measures of depression and anxiety (AYMH) 4.	1. Buccal DNA	1. Self-reported sociodemographic data	1. Investigating the effects of genetics and psychosocial environment on response to trauma in a case: control study of Syrian refugee and Jordanian non-refugee youth who were differentially exposed to war-related trauma.	Longitudinal case:control study	methylation. 3. Newborn methylation-PC1 showed a strong correlation with newborn birth weight. 4. All three maternal stressors were significantly correlated with each other, leading to material deprivation and mundane stress also being correlated with newborn methylation-PC1. 5. Maternal stress, NR3C1 methylation, and newborn birthweight were significantly correlated, but only in newborns and not in their mothers. 1. Identified direct genetic effect of 5-HTTLPR on human insecurity, which captures feelings of fear and insecurity in conflict- affected areas; high epression (HE) homozygotes had lower levels of insecurity. 2. Identified a GxE interaction of 5-HTTLPR with resilience and the perceived	1. Future studies should investigate additional genetic variants in a larger sample size, while maintaining the longitudinal study design.

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Mulligan et al. (under review)	Epigenetic signatures of intergenerational exposure to violence in three generations of Syrian refugees	Middle East (Jordan)	Three generations of Syrian refugee (n=48 families with 131 individuals)		Measures of behavioral and emotional mental health difficulties (SDQ) 5. Measures of PTSD symptoms (CRIES-8)	1. Buccal DNA		1. Assessing DNAm signatures of war-related violence by comparing germline, prenatal, and direct exposures to violence across three generations of Syrian refugees	Longitudinal study	stress scale, which is a generalized measure of psychosocial stress; the HE allele appeared protective (as a homozygote or heterozygote) and was associated with sharper reductions in PSS over time in individuals with high levels of resilience in contrast to low expression (LE) homozygotes. 1. Based on an epigenome-wide association study (EWAS), differential methylation regions (DMRs) were found to be associated with germline and direct exposure to violence, with 14 and 21 DMRs identified, respectively. 2. The largest difference in DNAm relative to unexposed controls was observed at a germline-associated DMR, cg01490163, with lower DNAm among those exposed in germline. 3. The highest DNAm levels	1. Future research should adopt a positive framework to investigate epigenetic signatures of adaptive mechanisms that underly the tenacity and successful expansion of human populations throughout evolution. 2. Future research should incorporate greater intergenerational analysis, involve a more diverse range of populations, and undertake causal validation of mechanistic pathways.

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Rodney and Mulligan, (2014)	A Biocultural Study of the Effects of Maternal Stress on Mother and Newborn Health in the Democratic Republic of Congo	Central Africa (Democratic Republic of Congo)	24 mother–newborn dyads		1. Stressful life experiences, chronic stress, and war-related trauma (oral history) 2. Measure of peri-natal stress	1. Blood DNA (maternal venous blood, placental tissue, and umbilical cord blood)	1. Sel-reported general and reproductive health 2. Anthropometric data (newborn gestational age and birthweight)	1. Examining the effects of maternal stress on maternal health and the possible intergenerational transmission of the effects of stress.	Cross-sectional study	relative to controls was observed at the germline-associated DMR, cg07462448, as well as two DMRs associated with direct exposure, cg14117527 and cg14832449. 4. Most DMRs showed the same directionality in DNAm change across germline, prenatal, and direct exposures, suggesting a common epigenetic response to violence. 5. Epigenetic age acceleration in association with prenatal exposure to violence in children was identified, highlighting the critical period of <i>in utero</i> development. 1. War stress have the most substantial effect, explaining 35% of the variance in birth weight. 2. Household education also explained a significant amount of birthweight variance and was protective against reduced birthweight, as	

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
Smeeth et al. (2023b)	Risk and resilience in Syrian refugee children: A multisystem analysis	Middle East (Lebanon)	Child-caregiver dyads from Syria (n=1359)	8–16 years	1. Measures of depressive symptoms (CES-DC) 2. Measures of PTSD symptoms (CPSS) 3. Measures of the parent-reported Strengths and Difficulties (SDQ) 4. Measures of the degree of	1. Stress-related biomarker data (hair CORT) 2. Saliva DNA	1. Self-reported sociodemographic data	1. Examining the potential of hair cortisol concentrations and polygenic scores for mental health outcomes to predict resilience to internalizing and externalizing mental health problems.	Longitudinal study, cross-sectional design of the analysis	well as age. 3. The placenta was hypomethylated relative to both maternal venous blood and newborn cord blood. 4. A negative correlation between mean global methylation and maternal war stress exposures. 5. A correlation between maternal war stress and methylation is seen only in maternal venous blood and not in newborn cord blood 1. War exposure was significantly associated with decreased resilience. 2. A significant association between elevated hair cortisol concentrations and decreased resilience 3. Polygenic risk for various mental health-related outcomes	1. Future studies on investigating combinations of markers such as hair cortisol and polygenic scores. 2. The importance of longitudinal studies in unpacking substantial complexities t (e. g., cortisol blunting; ethnicity-specific polygenic scores),

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Table 2 (continued)

Author	Title	Geographical area	Sample size	Age	Self-Reported Mental Health Measures	Physiological biomarker data and epigenetic data	Other	Purpose of study	Study design	Main findings	Future directions
					civilian exposure to war events (WEQ)					was not clearly associated with resilience. 4. Only one interaction, between hair cortisol and polygenic risk for depression (PGS001829), emerged as significant.	before biological markers can serve as credible screening tools in refugee contexts.

1991), without much work done in the Middle East or South Asia (Weaver et al., 2015).

4.2.3. Inflammation

Four studies have investigated associations between inflammation and war-related trauma. One (Ghandour et al., 2022) found positive associations between trauma exposure and IL-6, as well as an inverse correlation between depression and IL-6 among Iraqi and Syrian refugees. Anxiety scores showed a positive association with IL-8, while PTSD showed no significant associations with IL-6 or IL-8. The second study (Rohleder et al., 2004) observed an elevated production of the pro-inflammatory marker interleukin-6 (IL-6) in response to a lipopolysaccharide (LPS) challenge, along with higher cytokine producing capacity in peripheral blood monocytes, among Bosnian war refugees with PTSD, as compared to healthy individuals. Individuals in the PTSD group exhibited greater suppression of LPS-stimulated production of whole blood IL-6 and tumor necrosis factor-alpha (TNF-α) by dexamethasone when compared to the control group. The third study (Pitts et al., 2016) compared two groups: Bosnian war refugees undergoing treatment for their clinically diagnosed PTSD (n=52) and healthy German controls. The authors identified heightened glucocorticoid (GC) sensitivity concerning the production of inflammatory cytokines in whole blood, among refugees with PTSD in comparison to controls. They also observed an increased dampening effect of dexamethasone on the LPS-induced production of IL-6 in whole blood, a factor likely integral to the overall PTSD phenotype. The fourth study (Gettler et al., 2024) compared two refugee populations: recently settled refugees in Serbian asylum centers and refugees in Kakuma camp in Kenya who had experienced a prolonged stay. They found that refugees in Serbia had a lower likelihood of having elevated CRP compared to refugees at Kakuma, consistent with greater low-grade, chronic inflammation in the latter. The authors observed gender differences, with refugee men exhibiting lower likelihood of having elevated CRP compared to refugee women.

Heterogeneous findings could result from the fact that some individuals were undergoing mental health treatment. For example, two studies (Rohleder et al., 2004; Pitts et al., 2016) reported their participants undergoing psychological treatment at time of the study, whereas Ghandour et al.'s (2022) participants did not undergo treatment. Other confounding factors could include assay methodology, environmental factors, comorbidity, and body mass index.

4.2.4. Cardiovascular system

As one of the few studies examining the prevalence of hypertension and its associated mental health risk factors among female refugees in South Africa, Munezero and Tomita (2021) identified a noteworthy prevalence of hypertension in this group, with over 80% of participants exhibiting elevated blood pressure. The authors also found that exposure to adverse childhood experiences and depression correlated with hypertension, regardless of other factors, including food insecurity, alcohol and tobacco use, among others. The study underscored the complex interplay of mental health factors influencing the cardiovascular health of displaced populations.

4.2.5. Neuroplasticity

One study (Arnetz et al., 2020) stands out as one of the few examining neuroplasticity responses in the context of war trauma and refugee mental health, with data on brain derived neurotrophic growth factor (BDNF) and nerve growth factor (NGF). These two primary neurotrophins are crucial for neuroplasticity; levels are elevated in the blood of individuals diagnosed with PTSD (Zhang et al., 2014). BDNF has also been identified as a potential biological marker of positive health outcomes (resilience), where reduced BDNF activity is associated with increased sensitivity to stress (Notaras and van den Buuse, 2020). The authors found that the neuroplasticity index, characterized by increased levels of BDNF and NGF, exhibited correlations with PTSD, depression, and anxiety, suggesting a potential dysfunctional neuroplastic process in

Box 2

Challenges of research focusing on war-affected refugee populations.

Research design:

- Cross-sectional, longitudinal, experimental
- Sample and control group characteristics (age, sex, living conditions, mental health)
- Sample size and representativeness

War-related adversities:

- Trauma exposure (duration, severity, type)
- Timing between exposure and biomarker assessment

Biomarker standardization:

- Specimen collection (e.g., blood, saliva, hair, nail)
- Collection procedures (e.g., time of day, meal and caffeine consumption, smoking)

Mental health assessment:

- Psychometric tools measuring mental health burden
- Cultural relevance for refugee populations

Scientific reporting and ethical engagement:

- Ensuring ethical conduct and community engagement
- Detailing research procedures in the field, as well as laboratory and procedures

Syrian and Iraqi refugees in the USA. They also found connections between pre-displacement environmental exposure, PTSD, and anxiety, with Syrian refugees exhibiting generally poorer self-rated mental health. Another study (Thoma et al., 2012) examined diurnal profiles of the salivary enzyme alpha-amylase (sAA) in two groups: Bosnian war refugees with PTSD and healthy controls. This enzyme may constitute a convenient and non-invasive indicator of the sympathetic nervous system (SNS) activity (Rohleder and Nater, 2009). The authors found that the altered awakening sAA pattern in PTSD individuals, characterized by an increase instead of the expected decrease after awakening, potentially underscores the dysregulation of SNS. Findings indicated a correlation between sAA concentrations and the severity of trauma, along with symptoms of psychopathology and anxiety. Moreover, elevated wake-up responses in sAA were linked to increased GC sensitivity in the production of inflammatory cytokines.

On the one hand, Arnetz et al.'s (2020) examination of the neuroplasticity index presents a relatively novel perspective on the potential utility of neurotrophins as markers for dysfunctional neuroplastic processes. These neurotrophins may also play a role in shaping the mental health profiles of displaced individuals, particularly in the context of post-displacement experiences. On the other hand, Thoma et al.'s (2012) study sheds light on the nuanced diurnal patterns of sAA and its associations with trauma severity and psychopathology. This study also offered insights into sAA as a potential marker for sympathetic nervous system (SNS) dysregulation in individuals with PTSD. However, both relied on a cross-sectional design and a relatively limited sample size. Understanding the temporal sequence of events and the potential bidirectional influences between war trauma, neuroplasticity, and physiological responses thus remains elusive. Longitudinal research is here essential for gaining an understanding of the pathways through which war trauma and environmental exposures contribute to neuroplasticity

and SNS dysregulation.

4.3. Experimental evidence

Overview. The study presents experimental evidence from randomized control trials conducted with Syrian refugee and non-refugee youth in Jordan, evaluating the effectiveness of the *Advancing Adolescents* program in alleviating toxic stress. The interdisciplinary research team tracked physiological, epigenetic, cognitive, mental health, and resilience data over three time points to measure intervention impacts. Results showed that levels of physiological stress in war-affected adolescents were responsive to the intervention, with significant changes in cortisol production over time. The study provided biological and psychosocial data for a hard-to-reach, war-affected population group living in the Middle East, focusing attention on inter-individual differences in adolescent health and developmental trajectories. It demonstrated how to 'do science' through culturally-grounded methodologies and culturally-relevant approaches to risk and resilience (Panter-Brick, 2022). Findings also underscored the importance of interventions that can alleviate stress and boost social support in humanitarian settings.

Randomized control trial. Two publications offer experimental evidence from a randomized control trial conducted with Syrian refugee and non-refugee youth in Jordan (Dajani et al., 2018; Panter-Brick et al., 2020b). The team tracked physiological, epigenetic, cognitive, mental health, and resilience data at three time points (baseline, post-intervention, one-year follow-up) to evaluate the effectiveness of *Advancing Adolescents*, a program implemented by Mercy Corps to alleviate toxic stress in war-affected youth in the Middle East region. The interdisciplinary research team followed 12–18-year-olds (male and female; n=446 Syrians, n=371 Jordanians), randomized to intervention

Table 3
Population comparisons for biomarker data.^a

	EBV antibody titers (AU/mL)			Study description/population
	mean	SE or SD	N or Obs.	
Refugees in Serbia	166.67	16.46 (SE)	Obs. = 172	Data collection among refugees living in Serbia was carried out across three asylum centers (mean age: 29.71 years, 8.87 SD). Data collection included only refugees u of Kakuma 1, which is the original camp area in Kenya (mean age: 35.64 years, 10.99 SD) (Gettler et al., 2024).
Refugees in Kakuma	208.67	21.23 (SE)	Obs. = 139	
U.S. adults	171.49	1.42 (SE)	N = 11,050	Data from The National Longitudinal Study of Adolescent to Adult Health (Add Health): a large, nationally-representative study of U.S. adults (mean age: 28.37 years, range 24–32 years). (Dowd et al., 2014). ^b
Syrian refugee youth	144.07	142.04 (SD)	N = 309	Syrian refugee youth living in Jordan who fled the war in Syria (mean age: 14.33 years, 1.82 SD). (Panter-Brick et al., 2020b).
Latino immigrants to the U.S.	131.69	70.55 (SD)	N = 79	Latino immigrants residing in Oregon, U.S. (mean age: 35.0 years, 11.04 SD). (McClure et al. 2010).
Filipino adults	87.40	50.37 (SD)	N = 1248	Data from the Cebu Longitudinal Health and Nutrition Survey: a large multi-decade population-representative birth cohort study of Filipino adults (mean age: 21.47 years, SD 0.33). (Georgiev et al., 2016). ^b
	Fingernail cortisol (pg/mg)			Study description/population
	mean	SD		
Refugees in Serbia	4.43	7.90	Obs. = 225	Data collection among refugees living in Serbia was carried out across three asylum centers (mean age: 29.71 years, 8.87 SD). Data collection included only refugees u of Kakuma 1, which is the original camp area in Kenya (mean age: 35.64 years, 10.99 SD) (Gettler et al., 2024).
Refugees in Kakuma	2.59	6.84	Obs. = 148	
Japanese adults	5.90–6.80	1.80–3.70	N = 23	Japanese young adults (mean age: 22.60 years, SD 2.70). Participants provided nail samples monthly for eight months. (Izawa et al., 2021)
Chinese adults	5.41–5.65	1.63–1.88	N = 51	Chinese young adults (mean age: 20.02 years, SD 1.24). Participants provided nail samples twice thirty days apart. (Wu et al., 2018).
Congolese BaYaka adults	4.36	2.98	N = 49	BaYaka adults from a subsistence-level foraging society in northern Rep. of the Congo (mean age: 37.69 years, 11.77 SD). (Gettler et al., 2021)
Kenyan Daasanach adults	8.25	3.97	N = 155	Daasanach adults from a semi-nomadic pastoralist society in Kenya. (mean age: 43.10 years, 15.90 SD). (Rosinger et al., n.d.).
	Hair cortisol (pg/mg)			
	mean	SD		
Asylum seekers with PTSD	8.21	4.62	N = 32	Asylum seekers without PTSD (mean age 32.8 years, 6.9 SD) and with PTSD (mean age: 32.0 years, 7.6 SD) living in Germany for an average of 7 months. Control groups were healthy permanently settled Turkish immigrants (mean age: 24.3 years, 2.7 SD) and healthy nonimmigrant Germans as the reference group (mean age: 25.9 years, 3.8 SD) (Mewes et al., 2017).
Asylum seekers without PTSD	7.43	4.93	N = 24	
Turkish immigrants	4.72	2.23	N = 24	Women internally displaced Northern Iraq (mean age: 34.34 years, 11.71 SD) and Syrian women asylum seekers in Germany (mean age: 30.04 years, 5.25 SD) were compared to Kurdish immigrants and asylum seekers living in Germany, on average, for 18 years (mean age: 34.63 years, 9.39 SD) (Buchmüller et al., 2020).
Non-immigrant Germans	6.10	3.30	N = 28	
Internally displaced people	7.54	9.82	N = 14	Unaccompanied refugee minors predominately from Afghanistan and Syria living for 19 months in sheltered accommodation in Germany (mean age: 17.40 years 1.17 SD) (Sierau et al., 2019).
Asylum seekers	5.67	4.68	N = 37	
Kurdish immigrant	8.64	7.91	N = 38	Refugee mothers (mean age: 29.74 years 5.53 SD) and their children (mean age: 33.98 months old, 18.42 SD) living in Germany (Lembcke et al., 2020).
Unaccompanied refugee minors	1.56	0.21	N = 91	
Refugee children	1.82	0.90	N = 37	Data from a substudy of the German cohort of the European Prospective Investigation into Cancer and Nutrition (EPIC) study, a large multi-center prospective cohort study investigating associations between diet, lifestyle and chronic disease risk (mean age: 65.8 years, range 47 and 82 years old). (Feller et al., 2014).
Refugee mothers	1.49	0.68	N = 38	
German adults	35.1	32.8	N = 654	Data from the Whitehall II occupational cohort study of British civil servants (mean age 69.8, range: 59–83 years old). (Abell et al., 2016).
British adults	3.5	3.7	N = 3675	
	Elevated CRP ^c (% yes)			
	mean	SD		
Refugees in Serbia	20.6	-	Obs. = 165	Data collection among refugees living in Serbia was carried out across three asylum centers (mean age: 29.71 years, 8.87 SD). Data collection included only refugees u of Kakuma 1, which is the original camp area in Kenya (mean age: 35.64 years, 10.99 SD) (Gettler et al., 2024).
Refugees in Kakuma	36.6	-	Obs. = 145	
U.S. adults	35.4	-	N = 4349	Data from The National Longitudinal Study of Adolescent to Adult Health (Add Health): a large, nationally-representative study of U.S. adults (mean age: 37.9 years, range 33–44 years old). (McDade et al. 2021).
Filipino adults	7.3%	-	N = 1648	Data from the Cebu Longitudinal Health and Nutrition Survey: a large multi-decade population-representative birth cohort study of Filipino adults (mean age: 20.9 years, SD 0.30). (McDade et al., 2009)

^a Adapted from Gettler et al. (2024) Refugee health and physiological profiles in transitional settlements in Serbia and Kenya: Comparative evidence for effects of gender and social support. PNEC 165, 107024.

^b The values from Dowd et al. (2014), Georgiev et al. (2016) and McClure et al. (2010) were converted using the formula provided in Eick et al. (2016). The SE value provided in Table 3 is directly from Dowd et al. (2014), and SE values for EBV antibody titer data from refugees in Serbia and Kenya are reported for comparison (Gettler et al. 2024).

^c Elevated CRP: >3 mg/L and <10 mg/L.

and control groups.

The study was able to measure changes in mental health and physiological trajectories of Syrian and Jordanian individuals over time. Analyses used growth mixture modeling to identify cortisol trajectories over time, identifying hypersecretion, medium secretion, and hyposecretion, in 8.6%, 78%, and 2.6% of the cohort, respectively (Panter-Brick et al., 2020b). Experiencing insecurity or fear was predictive of increased cortisol production. Stronger posttraumatic stress reactions were associated with a pattern of within-individual cortisol dysregulation and with medium secretion. Thus for each additional symptom reported on the PTSD screening instrument, adolescents were 0.07 times less likely to show hyposecretion (Dajani et al., 2018).

The research team also demonstrated intervention impacts: levels of physiological stress in war-affected adolescents were malleable and responsive to *Advancing Adolescents*, an 8-week program based on stress attunement (mindfulness) and dedicated support. As a result of participating in the Mercy Corps program, Syrian and Jordanian youth normalized cortisol production over time. Specifically, the intervention group - as compared to controls - showed a significant lowering of cortisol production, equivalent to a one-third (33%) reduction in HCC levels (Dajani et al., 2018; Panter-Brick et al., 2020b). Remarkably, the intervention elevated cortisol levels in the hyposecretion group, who showed a blunted response to environmental challenges, as well as reduced cortisol levels in the hypersecretion group, who showed chronic hyper-arousal in adversity. Such findings offered Mercy Corps valuable reasons for developing humanitarian practice and humanitarian donors for funding youth programs that prioritized stress-attunement and social support in the Middle East region (Panter-Brick, 2023).

The study showed what could be accomplished in humanitarian settings in terms of implementing a research design that reflected both scientific rigor and cultural engagement (Panter-Brick et al., 2020a; Panter-Brick, 2021). In terms of cultural engagement with a view to ensure local relevance, for example, the team developed the Arabic version of the Child and Youth Resilience Measure (CYRM) because youth had asked the research team to call attention to their strengths, not just mental health vulnerabilities, in response to adversity. In the field, female and male hairdressers from the local area were employed to facilitate hair cortisol data collection, allowing for time to build rapport and for opportunities to offer professional, complimentary haircuts. The research was highlighted as an exemplary case study of interdisciplinary collaboration and partnerships in humanitarian crises by its U.K. funding body (Elrha, 2023a,b). It was also chosen as a case analysis for humanitarian health research by the Fogarty International Center of the U. S. National Institutes of Health (Mistry et al., 2021).

4.4. Observational studies

Overview. This body of work shows a mix of evidence, suggesting that the pathophysiology of neuroendocrine responses and inflammation may not follow a straightforward trajectory with respect to development of different trauma-related disorders in *refugee* populations. Inter-individual differences among refugee populations will stem from both potential genetic and environmental influences. We note that studies are heterogeneous across exposures of past trauma, chronic disease burden, duration of forced displacement, and other war-related and resettlement adversities. We also note heterogeneity in sample composition as well as sample size; for instance, Smeeth et al. (2023a) did not use a comparison or control group, Yamin et al. (2021) focused exclusively on male refugees, and Grasser et al.'s (2020) included just 36 adult refugees resettled in the USA.

4.4.1. Neuroendocrine system

Two longitudinal study were able to follow a cohort of 1591 Syrian child-caregiver pairs, recruited from informal tented settlements, in Lebanon. It is known as the BIOPATH (Biological Pathways of Risk and Resilience in Syrian Refugee Children) study. The research team

(Smeeth et al., 2023a) reported that older children exhibited both elevated HCC and increased exposure to war events, and found a correlation elevated HCC and severity of PTSD symptoms. The other study (May et al., 2024) supported the correlation between elevated hair cortisol concentrations (HCC) and increased PTSD symptoms. Additionally, the research team found that apart from cortisol, adolescents exhibited higher levels of two other hormones, DHEA and testosterone, compared to younger children, as well as higher levels in individuals with black hair compared to those with brown hair, with a trend of decline observed over time post-departure from Syria. The study also identified sex differences, with females displaying significantly elevated cortisol levels but lower concentrations of testosterone and DHEA in comparison to males.

However, the authors could not definitively conclude whether war exposure led to lasting changes in the HPA axis: such associations might be confounded or mediated by environmental factors, parent mental health, caregiving, or physical activity. There was no control group of child-caregiver pairs, because Syrian refugee families had all been exposed to war and stressful living conditions (Smeeth et al., 2023a; May et al., 2024).

4.4.2. Inflammation

We detail three studies that have examined psychopathology and inflammatory processes in refugees. The first (Grasser et al., 2020) reported that lower CRP basal levels were predictive of depressive and anxiety symptoms among Iraqi and Syrian refugees resettled to the USA. As for pro-inflammatory cytokine interleukin-1 beta (IL-1 β), lower concentrations were associated with higher PTSD symptomology, whereas higher IL-1 β expression predicted symptoms of anxiety and depression. The second study (Yamin et al., 2021) involved a group of male Iraqi refugees, recruited from a longitudinal study (Arnetz et al., 2014). Conversely, it reported that higher CRP levels were associated with PTSD and depressive symptoms at the 2-year follow-up. Conceptualizing acculturation as a potentially adaptive process, Yamin et al. (2021) identified associations between higher levels of acculturation, lower CRP levels, and lower depressive symptoms, though no associations with PTSD symptoms. However, it is likely that experiences of acculturation, forced displacement, and resettlement are very different from one refugee group to the next, as refugees often face poor mental health, social discrimination, poverty, and diminished social support.

The third study (Söndergaard et al., 2004) stands out as one of the few that investigated dehydroepiandrosterone sulfate (DHEA-s). The authors measured Serum amyloid A (SAA), CRP and DHEA-s among Iraqi refugees with and without PTSD. Serum SAA and high-sensitive CRP were lower in PTSD than in non-PTSD refugees. SAA and CRP levels exhibited an inverse correlation with serum DHEA-s, but not with cortisol. In individuals with PTSD at baseline, there was a negative association between the change in self-rated PTSD symptoms and the change in log-transformed SAA, indicating that an increase in PTSD intensity was linked to a decrease in SAA. No discernible patterns were observed regarding CRP.

4.5. Genomic studies

Overview. This body of work examines the impact of early-life adversities on human biological variation and the intergenerational effects of maternal trauma. The developmental origins of health and disease (DOHaD) framework, which suggests that early childhood stressors disproportionately impact adult health (Gluckman et al., 2010), has served as a guiding framework for these studies (e.g., Clukay et al., 2019a, 2019b; Mulligan et al., 2022). While contributing to our understanding of the interplay between genetic and environmental factors in refugee populations, these studies offer mixed findings in terms of maternal stress, trauma, and DNAm patterns. This underscores the necessity of broadening the scope of research to a systems-view understanding of developmental pathways, and specifically, to understanding

the biological mechanisms involved in inter-individual responses to early childhood adversities and the intergenerational transmission of stress and trauma. Both initial work (Hjort et al., 2021; Mulligan et al., 2012) and subsequent publications (Rodney and Mulligan, 2014; Kertes et al., 2016; Kertes et al., 2017) share common challenges, such as small sample sizes and reliance on a cross-sectional study design. For instance, studies in the Democratic Republic of Congo (Mulligan et al., 2012; Rodney and Mulligan, 2014; Kertes et al., 2016, 2017) included 24 mother-newborn dyads to evaluate intergenerational impacts.

4.5.1. Refugee youth

We captured four studies that focused on refugee youth. The first is the longitudinal BIOPATH study (Smeeth et al., 2023b), notable for its large sample of Syrian children ($n=1359$). Here, the research team investigated whether HCC and polygenic scores were predictive of risk of poor mental health outcomes and resilience. In logistic regression models, elevated HCC were associated with fewer mental health problems, even after accounting for war exposure levels. There was a noteworthy interaction between HCC and polygenic scores for depression, suggesting that children predisposed to depression faced a higher risk of mental health problems when they had elevated cortisol production.

The second study (Javanbakht et al., 2021) compared two groups of child-caregiver dyads ($n=1359$): war-traumatized Syrian and Iraqi refugees and non-traumatized Arab immigrants, both resettled to the USA. Data for refugee participants were gathered at three different time points, one to three years following resettlement in the USA, while data for Arab immigrants were collected only once, within 3–5 years after immigration. Various biomarkers, including skin conductance responses (SCR) as indicators of autonomic response, saliva samples for genetic and epigenetic analyses, and data pertaining to the social and environmental context, such as family structure, resources, and neighborhood quality, were collected. In their initial analyses, Javanbakht et al. (2021) observed elevated anxiety levels in refugee children, along with heightened PTSD symptoms and anxiety in their parents. Furthermore, they noted a substantial responsiveness of skin conductance responses to trauma narratives in refugee children. There was a robust correlation between DNAm age and chronological age and a substantial responsiveness of skin conductance (SC) to children's trauma narratives. Moreover, half of the top 10 CpG sites, identified with a false discovery rate below 5%, were found to be associated with elevated anxiety in refugee children.

The third study (Clukay et al., 2019a) examined genetic variants of monoamine oxidase A (MAOA) to assess the impacts of early life trauma exposure in 12–18-year-old Syrian refugee youth ($n=399$, 45% girls). It assessed MAOA, analyzed from buccal swabs, in relation to perceived stress (PSS) and resilience (CYRM) data. The authors found an association of the MAOA variant with PSS scores over time. They noted an additive effect whereby the sharpest PSS reduction occurred in low-activity MAOA males with either low trauma exposure or high CYRM levels. This establishes associations between MAOA variants, resilience, and perceived psychosocial stress within this refugee group. The authors then explored genetic variants in three additional candidate genes (FAAH, the 5-HTTLPR region of SLC6A4, and BDNF) for potential associations with psychosocial stress and mental health outcomes (Clukay et al., 2019b). They found no significant associations between these additional candidate genes and mental health outcomes. Utilizing the case:control design of randomized trial, Mulligan et al. (2022) investigated genetic variants in two genes, 5-HTTLPR and an associated A/G SNP, rs25531, in the serotonin transporter gene 5-HTT, and the Val158Met SNP in catechol-O-methyltransferase (COMT). They identified genetic and environmental factors, as well as GxE interactions, associated with three measures of psychosocial stress (PTSS, perceived stress, and human insecurity). They also identified two GxE interactions with the tested loci, 5-HTTLPR and COMT, in relation to resilience and its association with perceived stress and PTSS, respectively. In both cases, the 5-HTTLPR HE variant and the COMT Val variant exhibited a

protective effect that manifested only in individuals with high levels of resilience. Notably, these three publications underscored the importance of ethically collecting data, including biological samples, in war-affected populations.

The fourth study (Mattonet et al., 2023) used a cross-sectional design to investigate the associations between war trauma, PTSD, and changes in DNAm patterns in Burundian families ($n=317$ participants) living in refugee camps in Tanzania. The epigenome-wide association study (EWAS), analyses on the Weighted Gene Co-expression Network Analysis (WGCNA) modules, and post-hoc sensitivity analyses revealed that neither the war-related trauma load nor the severity of PTSD symptoms exhibited significant associations with the epigenome-wide DNAm levels of the retrieved CpGs. This held true for both children and caregivers, as well as for the module's EigenCpGs from WGCNA.

4.5.2. Intergenerational effects

Three studies focused on intergenerational effects of maternal mental health on child wellbeing and development. This first (Hjort et al., 2021) pioneered the exploration of DNAm and cortisol levels in the offspring of mothers who endured PTSD arising from war-related sexual trauma during the Kosovo war. The authors found evidence that children born to mothers with PTSD during pregnancy had a higher likelihood of elevated cortisol levels, as compared to children born to mothers without pregnancy-related PTSD. Using multivariate linear regression analysis, adjusting for mother's age, offspring gender and age, batch effect, and cell composition, the authors identified 12,337 differentially methylated CpG sites ($p < 0.05$) with a $>1.0\%$ difference in DNA methylation between PTSD and non-PTSD exposed offspring. However, none remained significant after false discovery rate correction for multiple testing. The largest mean methylation difference between the groups was 12.0%, with approximately half (51.6%) of the differentially methylated CpGs showing higher DNA methylation levels among offspring exposed to PTSD during pregnancy compared to non-exposed offspring. Furthermore, offspring exposed to maternal PTSD during pregnancy compared to those who were not had differential DNA methylation degree at seven candidate genes (*NR3C1*, *NR3C2*, *HTR3A*, *SLC6A4*, *OXTR*, *FKBP5*, *BDNF*). These children also exhibited higher cortisol levels and differential methylation at candidate genes (*NR3C1*, *HTR3A*, and *BDNF*); yet there were no methylation differences that attained significance at the epigenome-wide corrected levels.

The second study (Mulligan et al., 2012) involved 24 mother-newborn dyads from the war-affected region of eastern Democratic Republic of Congo (DRC). The authors found correlations between maternal stressors (war trauma, material deprivation, and daily stressors) and the newborn birth weight, with war exposure having the most substantial effect. Regarding the effect of stress on methylation patterns, differences were observed in the correlation of *NR3C1* methylation data and maternal stressors, particularly when analyzing mothers and newborns separately. Maternal stress, *NR3C1* methylation, and birth weight were correlated, but only in newborns and not in their mothers. The strongest correlation was found between war trauma and newborn methylation. Moreover, newborn methylation-PC1 exhibited a robust correlation with birth weight. Material deprivation and mundane stress were also correlated with newborn methylation-PC1.

Building upon these findings in DRC, Rodney and Mulligan (2014) found a placental hypomethylation, relative to maternal venous blood and newborn cord blood. Noting that increased maternal stress was associated with elevated fetal *NR3C1* methylation and, subsequently, reduced fetal birth weight, the researchers measured placental hypomethylation relative to maternal venous blood and newborn cord blood. The inverse correlation between global methylation and maternal war exposures was specifically observed in maternal venous blood, not in newborn cord blood. As for the inverse association between war exposure and birth weight, the study highlighted the role of protective effects such as household education.

In another DRC publication, Kertes et al. (2016) tested for

associations between prenatal stress and DNAm at sites located in the multiple genes (*CRH*, *CRHBP*, *NR3C1*, and *FKBP5*). A total of 18 CpG sites were significantly associated with chronic stress or war trauma in one or more biological tissue. Among the CpG sites associated with prenatal stress, 11 were linked to chronic stress, and 14 were associated with war trauma, with six sites predicted by both stressors. Eight CpG sites, all associated with war trauma, survived FDR correction with moderate confidence. Considering CpG sites surviving FDR correction, those with prior evidence of association, or located at a transcription factor binding (TFB) region, chronic stress predicted methylation in *CRH* and *NR3C1*, accounting for 16%–25% of the variance. Chronic stress and war trauma made impacts on HPA axis gene methylation, with notable effects observed at a transcription factor binding (TFB) sites across all tested target genes. For placenta and maternal blood, effects were observed in all four genes, while in cord blood, significant associations with stressors were found in *CRH* and *NR3C1* only. Three potentially relevant *CRH* sites were identified in cord blood after false discovery rate correction. In placenta, reduced methylation at three *NR3C1* sites was significantly associated with war trauma. In maternal blood, one *NR3C1* site showed significant methylation associated with war trauma and nominal association with chronic stress. One *FKBP5* site showed higher methylation with chronic stress and war trauma in placenta and with chronic stress in maternal blood. The methylation of multiple CpG sites in *NR3C1* and *CRH*, all situated at TFB sites, was linked to birth weight.

In the third publication, Kertes et al. (2017) looked at BDNF gene methylation. In assessing methylation levels at 67 CpG sites across the BDNF gene in maternal venous blood, umbilical cord blood, and placental tissues, they found that 23 CpG sites were significantly associated with chronic stress or war trauma, with 12 surviving false discovery rate (FDR) correction, predominantly linked to war trauma. Chronic stress predicted methylation at six sites, explaining 13–26% of the variance, while war trauma was a predictor for methylation at 16 sites, explaining 12–34% of the variance. Maternal blood, placenta, and cord blood exhibited distinct associations between methylation levels and war trauma or chronic stress at specific CpG sites. In addition, sodium bisulfite sequencing data targeting the exon 4 promoter in placental tissue showed uniformly low methylation levels across CpG sites. In placenta, three CpG sites retained significant associations with war trauma after FDR correction, and in cord blood, two CpG sites showed significant associations with stressors after FDR correction.

The third study (Mulligan et al., under review) employed a novel three-generation approach to investigate the impact of violence exposure on hard-to-reach refugee communities. The research team assessed DNAm signatures of war-related violence across three generations of Syrian refugees in Jordan ($n = 48$ families with 131 individuals). They compared families where either a pregnant grandmother or a pregnant

mother experienced war violence and included a control group of Syrians without war exposure. In conducting an EWAS, the authors found 14 differentially methylated regions (DMRs) associated with germline exposure to violence and 21 DMRs associated with direct violence exposure. Specifically, the largest difference in DNAm relative to unexposed controls was observed at a germline-associated DMR, cg01490163, with lower DNAm among those exposed in germline. The highest DNAm levels relative to controls was observed at the germline-associated DMR, cg07462448, as well as two DMRs associated with direct exposure, cg14117527 and cg14832449. These findings suggest that DNAm could serve as a mechanism for transmitting information about trauma across generations. Most DMRs exhibited consistent DNAm changes across germline, prenatal, and direct exposures, indicating a shared epigenetic response to violence. Additionally, the study identified epigenetic age acceleration linked to prenatal violence exposure in children, underscoring the importance of the prenatal period in development.

5. Discussion

In this paper, we reviewed current knowledge on the biological signatures of war and forced migration. We make four key observations (Box 3). First, genomic and physiological stress biomarkers are valuable indicators for measuring and understanding human biological variation in the wake of violence, stress, and trauma. Epigenetic modifications, such as DNA methylation patterns, can be indicative of how environmental stressors influence gene expression and contribute to long-term physiological changes. Similarly, stress biomarkers, like cortisol levels, measure physiological responses to challenges and can offer biological evidence of the signatures of war and displacement. By implication, these biological markers are useful for providing a better understanding of why and how the biological and developmental signatures of war and forced displacement are so variable within and across population groups.

Biological markers provide a window on the complexity of stress response systems and the developmental processes that matter for child, adult, and intergenerational health outcomes. They help us understand the mechanisms by which risk and resilience pathways leave epigenetic and physiological signatures on the body, with developmental implications for young children and long-term health consequences for adults (Southwick et al., 2014). Other promising approaches include using biomarkers from tooth tissue (Dunn et al., 2022) and measuring the Pediatric-Buccal-Epigenetic (PedBE) clock (McGill et al., 2022) to track prenatal stress, early-life adversity, and physical and/or mental health risks across the lifespan. Future observational studies, for instance, could investigate how these markers interrelate with other biomarkers of developmental disruption, such as neural markers and stress

Box 3

Key observations of current knowledge on the physiological and genomic signatures of war and forced migration.

1. Genomic and physiological stress biomarkers are valuable indicators for understanding biological variation in the wake of violence, stress, and trauma. Yet they are not straightforward indicators of health vulnerability; they must be situated in developmental, economic, and sociocultural context.
2. Experimental and longitudinal evidence is crucial to unravel causal relationships between human epigenetics, physiology, human experiences and war-related adversities. Intervention studies can inform the development of effective strategies for health promotion in forcibly displaced populations.
3. Research with war-affected populations presents numerous ethical challenges. Collaborations that foster local engagement are critical to cultivate shared ownership of data and sustainable research initiatives.
4. It is challenging to engage with theory, report interdisciplinary data, and address methodological challenges in peer-reviewed articles. There is room for improvement in explaining how biological research is done on the ground and why collaborations work.

reactivity (Davis et al., 2020).

Yet biomarkers are not straightforward indicators of adversity, biological adaptability, or health vulnerability (Panter-Brick et al., 2020b). This is because the ways individuals perceive and respond to social situations, as well as the support provided by caregivers to their children, can significantly influence and shape their biological responses to adverse challenges. Similarly, the distribution of life-sustaining resources by local communities and national governments may change health and developmental pathways, in ways that cannot necessarily be captured by biomarker measurements. We know that individual, family, social, economic, and political domains influence the ways human grow up and respond to adverse challenges, leading to differences in psychoneuroendocrinology and, by extension, to differential health outcomes. For example, at the individual level, age of exposure matters to developmental and physiological trajectories in the wake of war and forced displacement, whilst family-level social support, access to healthcare, and precarity related to one's legal status in a host country, contribute to heterogenous outcomes for biological wellbeing. Importantly, rather than taking stress and biological markers at face value, future work needs to embrace the careful description of inter-generational, sociocultural, economic, and political experiences in war-related contexts.

Second, we need to improve research designs, moving beyond cross-sectional analyses of human biology and war-related adversity, striving for experimental and longitudinal evidence to unravel causal and developmental relationships between epigenetics, physiology, and war experiences. For example, well-conducted experimental trials, tracking the biological signatures of health-promoting interventions, can help us understand the malleability of biological responses and the extent to which, and for whom, stress and trauma are long-lasting. Of course, selecting an appropriate control group and a range of biomarkers are crucial decisions in the study design. In this review, we noted that comparisons are often drawn between refugees and individuals without PTSD, without careful attention to potentially important variables such as socio-economic stressors and sources of social support (Table 3). Similarly, a glaring limitation in the studies under review lies in their small sample sizes. To overcome this impediment and fully acknowledge individual differences, future research must aim for larger sample sizes. The choice of biomarkers must be contingent upon the research question, identified evidence gaps, logistical and cultural feasibility, and theoretical and practical implications. To-date, biomarker research with refugee populations has provided valuable insights into the short-term and long-term biological signatures of war-related adversities. Future work, employing mixed-methods, longitudinal, and/or experimental study designs will yield a more grounded understanding on causal pathways and mechanisms underlying inter-individual differences in human health and development.

Third, research with war-affected populations presents numerous ethical challenges. Scholars bear a profound ethical responsibility to be cognizant of methodological challenges, to engage ethically with local communities, to partner with relevant institutions, and to conduct themselves in ways that do no harm (Taki and de Melo-Martin, 2021; Panter-Brick, 2021). Funding agencies have begun to provide concrete examples of ethical engagement in humanitarian contexts (Elrha, 2023), while scholars are increasingly explicit in highlighting ethical concerns when setting global priorities for humanitarian research and practice (Tol et al., 2020). This commitment underscores the need for researchers to share their study protocols and reassess research practices in light of changing circumstances and community needs in war or refugee settings. To accomplish this, biological work requires a robust collaboration with in-country partners and local communities. For example, taking time to engage young people and their families in community events, to explain the novel science of biomarkers, and employing local hairdressers to collect hair samples for HCC analysis are ways to make the collection of biomarkers culturally-compelling (Panter-Brick, 2022). Collaborations that foster local engagement are also critical to cultivate

shared ownership of data and sustainable research initiatives.

Fourth, we observe a challenge in the scientific reporting of interdisciplinary data. For example many publications lacked substantial engagement with theoretical frameworks. While pioneering the usage of biomarkers in refugee populations, they often omitted to relate empirical findings with a consideration of theoretical perspectives. The limited engagement with theory poses a challenge to the broader theoretical synthesis of the research in this field. Furthermore, the structure of peer-reviewed articles frequently leaves little room to explain methodological challenges in the field and strategies adopted to foster strong, equitable partnerships with war-affected families, community leaders, or scientific partners. There is room for improvement in detailing how biological research is done on the ground and why collaborations matter. For example, longitudinal work with forcibly displaced people must contend with potentially high attrition rates over time, given that refugees are often highly mobile populations. Detailing challenges on the ground, such as the choice of control groups and respect for social norms regarding the human body, would contribute to a richer understanding of research processes and impacts.

Finally, this review has offered compelling evidence of the positive effects of psychosocial interventions and community-based approaches on the health and physiology of forcibly displaced populations. For instance, randomized control trials have demonstrated the effectiveness of interventions that prioritize stress-attunement and social support in alleviating the physiological effects of adversity (Panter-Brick et al., 2020a). These findings bear significant practical implications for policymakers, practitioners, and humanitarian organizations actively involved in addressing the health needs of forcibly displaced communities. Biomarker research has proven its value for humanitarian and governmental programs. For example, in the Middle East region, humanitarian actors were able to leverage international funding and state support for adolescent mental health programming, because biomarkers conclusively demonstrated that participation in a stress-attunement intervention would beneficially regulate cortisol trajectories in Syrian refugees and Jordanian youth (Panter-Brick, 2022).

Our review has two main limitations. First, it includes a small number of studies, published in English, that offer data on human physiology, genetics, and epigenetics in the context of war and forced migration. It is worth noting the absence of biological research conducted with internally displaced persons, despite their global significance. Second, it does not include all published work on genetic or neurological data in refugees. We did, however, include epigenetic studies, because these offer novel insights into developmental processes and intergenerational trauma.

6. Conclusion

To critically synthesize knowledge to-date can help identify key issues and directions for future research. We hope that this review can pave the way for a comprehensive understanding of the malleability of human biological responses to war-related adversities, providing practical knowledge on how to develop ethically-grounded, scientifically-informed research that can, in turn, inform practice and policy.

CRedit authorship contribution statement

Catherine Panter-Brick: Writing – review & editing, Writing – original draft, Conceptualization. **Jelena Jankovic-Rankovic:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

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